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Information Technology - Service management - 信息技术 - 服务管理 -

Part 2: 第二部分

Code of Practice 实用规则

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Information Technology - Service management -

信息技术 – 服务管理 –

Part 2: 第二部分

Code of practice 使用规则

前言 FOREWORD

ISO（国际标准化组织）和IEC（国际电工委员会）构成世界范围内的标准化专门体系。国家机构作为ISO或IEC的成员，通过由处理特定技术领域的相应组织所建立的技术委员会参与开发国际标准。ISO和IEC的技术委员会在共同关心的领域合作。其他的国际性组织，官方或非官方的，也与ISO和IEC联系，参与部分工作。在信息技术领域，ISO/IEC建立了联合技术委员会，ISO/IEC JTC 1。

国际标准依据ISO/IEC指南第2部分起草。

联合技术委员会的主要任务是准备国际标准。联合技术委员会接受的国际标准草案将提交给国家机构进行投票。成为国际标准公开发布，则需要至少75%的国家机构投赞成票。

ISO (the International Organization for Standardization) and IEC (the International Electrotechnical Commission) form the specialized system for worldwide standardization. National bodies that are members of ISO or IEC participate in the development of International Standards through technical committees established by the respective organization to deal with particular fields of technical activity. ISO and IEC technical committees collaborate in fields of mutual interest. Other international organizations, governmental and non-governmental, in liaison with ISO and IEC, also take part in the work. In the field of information technology, ISO and IEC have established a joint technical committee, ISO/IEC JTC 1.

International Standards are drafted in accordance with the rules given in the ISO/IEC Directives, Part 2.

The main task of the joint technical committee is to prepare International Standards. Draft International Standards adopted by the joint technical committee are circulated to national bodies for voting. Publication as an International Standard requires approval by at least 75 % of the national

应注意本标准的某些内容可能涉及专利。ISO和IEC不负责识别任何专利。

bodies casting a vote.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. ISO and IEC shall not be held responsible for identifying any or all such patent rights.

ISO/IEC 20000-2由BSI 起草 (BS15000-2)，被ISO/IEC JTC1 信息技术联合技术委员会按“快速跟进程序”所接纳，并得到了ISO和IEC的国家机构批准。

ISO/IEC 20000-2 was prepared by BSI (as BS 15000-2) and was adopted, under a special “fast-track procedure”, by Joint Technical Committee ISO/IEC JTC 1, Information technology, in parallel with its approval by national bodies of ISO and IEC.

ISO/IEC 20000由以下两部分组成，采用相同的名称《信息技术 – 服务管理》

ISO/IEC 20000 consists of the following parts, under the general title Information technology — Service management:

- 第一部分：规范
- 第二部分：实用规则

- Part 1: Specification
- Part 2: Code of practice

介绍 INTRODUCTION

ISO/IEC 20000标准的本部分，作为实用规则，采用指南和建议的方式。不应该作为规范来引用，并且特别要注意不要对符合性的宣称构成误导。

As a code of practice, this part of ISO/IEC 20000 takes the form of guidance and recommendations. It should not be quoted as if it were a specification and particular care should be taken to ensure that claims of compliance are not misleading.

ISO/IEC 20000标准的本部分的利用应该配合ISO/IEC 20000-1中与实

This part of ISO/IEC 20000 should be used in conjunction with ISO/IEC 20000-1, the specification associated with this

用惯例相关的规范进行。

code of practice.

实施本标准的前提条件是实施人员具备相应的资质和能力。作为国际标准不声称包括一个合同所有必要的条款。标准的使用者对其正确的利用负责。

It is assumed that the execution of the provisions of this part of ISO/IEC 20000 is entrusted to appropriately qualified and competent people. An International Standard does not purport to include all necessary provisions of a contract. Users of International Standards are responsible for their correct application.

符合标准本身并不获得法律义务的豁免。

Compliance with an International Standard does not of itself confer immunity from legal obligations

ISO/IEC 20000的本部分描述了ISO/IEC 20000-1范围内服务管理过程的最佳惯例。

This part of ISO/IEC 20000 describes the best practices for service management processes within the scope of ISO/IEC 20000-1.

服务交付的重要性不断提高，同时客户的要求更好的便利服务（在最低成本条件下）以满足其业务需求。而且意识到服务和服务管理是帮助企业获得收益和成本效益最佳的基础。

Service delivery grows in importance, as customers require increasingly advanced facilities (at minimum cost) to meet their business needs. It also recognizes that services and service management are essential to helping organizations generate revenue and be cost-effective.

ISO/IEC 20000-1描述了服务管理规范，应该结合ISO/IEC 20000的本部分进行阅读。

ISO/IEC 20000-1 is a specification for service management and should be read in conjunction with this part of ISO/IEC 20000.

ISO/IEC 20000系列标准使得服务提供方理解怎样提高向其内部和外部客户交付的服务的质量。

The ISO/IEC 20000 series enables service providers to understand how to enhance the quality of service delivered to their customers, both internal and external.

随着对服务支持依赖的和可利用技

With the increasing dependencies in support services and

术的增长，服务提供方能够致力于维护更高层次的客户服务。反应式的工作模式是，在计划、培训、评审、调查以及与客户共同工作方面花很少的时间。这是未能利用结构化的、预防式的工作模式的结果。

而这些服务提供商被要求提高服务质量、降低成本、更便利和更快的响应。有效的服务管理将向客户提交更好的服务和获得更高的客户满意度。

the diverse range of technologies available, service providers can struggle to maintain high levels of customer service. Working reactively, they spend too little time planning, training, reviewing, investigating, and working with customers. The result is a failure to adopt structured, proactive working practices.

Those same service providers are being asked for improved quality, lower costs, greater flexibility, and faster response to customers. Effective service management delivers high levels of customer service and customer satisfaction.

The ISO/IEC 20000 series draws a distinction between the best practices of processes, which are independent of organizational form or size and organizational names and structures. The ISO/IEC 20000 series applies to both large and small service providers, and the requirements for best practice service management processes do not change according to the organizational form which provides the management framework within which processes are followed.

1. 范围 SCOPE

本标准描述了有关IT服务管理过程的业内认可的质量标准。这些服务管理流程在商定的资源水平下，交付最佳可接受服务以满足客户业务需求，

This part of ISO/IEC 20000 represents an industry consensus on quality standards for IT service management processes. These service management processes deliver the best possible service to meet a customer's business

即专业、经济的服务并且风险得到了理解和管理。

在同一过程，或不同过程/功能组（岗位）之间使用不同的术语，可能给新的管理者在服务管理主题上带来混乱。在流程和功能组（岗位名称）使用新的管理都混淆服务管理的目标。如果不能理解术语，将给建立有效的流程带来障碍。理解ISO20000的术语可以带来可见的显著收益。本标准推荐服务提供商采用通用的术语，并对服务管理采取一致的方法。这将成为服务改进的一般基础。同时也为服务管理工具的供应商提供了框架。

作为给予过程的实用规则本标准，并不试图用于产品评估。然而，开发服务管理工具、产品和系统的组织可以规范和使用规则来帮助其开发服务管理工具、产品和系统，以支持服务管理的最佳实践。

本标准为审核提供指南，并帮助服务提供商计划服务改进或迎接ISO/IEC20000-1的审核。

ISO/IEC20000-1规定了多个紧密相关的服务过程，如图1所示。

needs within agreed resource levels, i.e. service that is professional, cost-effective and with risks which are understood and managed.

The variety of terms used for the same process, and between processes and functional groups (and job titles) can make the subject of service management confusing to the new manager. Failure to understand the terminology can be a barrier to establishing effective processes. Understanding the terminology is a tangible and significant benefit from ISO/IEC 20000. This part of ISO/IEC 20000 recommends that service providers should adopt common terminology and a more consistent approach to service management. It gives a common basis for improvements in services. It also provides a framework for use by suppliers of service management tools.

As a process based standard this code of practice is not intended for product assessment. However, organizations developing service management tools, products and systems may use both the specification and the code of practice to help them develop tools, products and systems that support best practice service management.

This part of ISO/IEC 20000 provides guidance to auditors and offers assistance to service providers planning service improvements or to be audited against ISO/IEC 20000-1.

ISO/IEC 20000-1 specifies a number of related service management processes as shown in Figure 1.

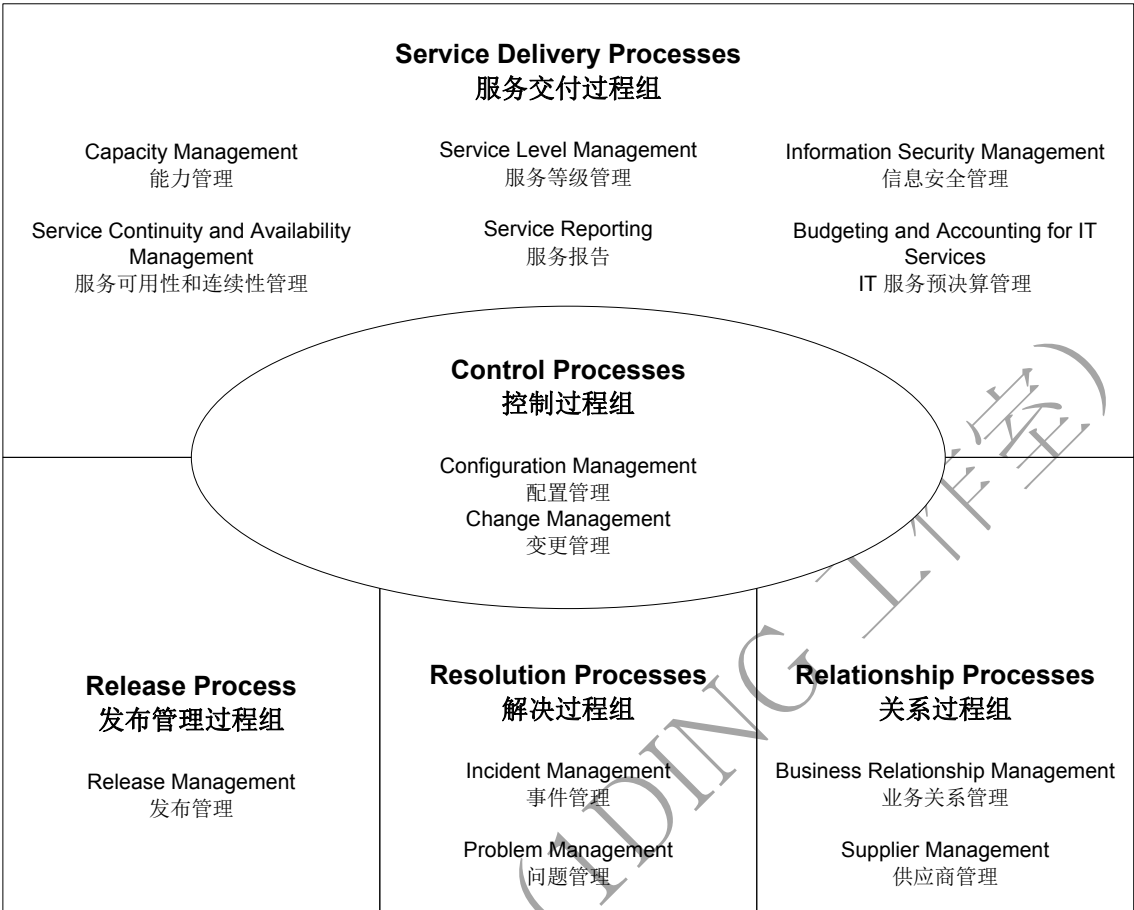


Figure 1 Service management processes 图 1 服务管理过程

2. 术语及定义 TERMS AND DEFINITIONS

本标准采用ISO/IEC 20000-1术语和定义。

For the purposes of this document, the terms and definitions given in ISO/IEC 20000-1 apply.

3. 管理体系 THE MANAGEMENT SYSTEM

目标：提供包括策略和架构的管理体系，使有效的管理和实施IT服务。

Objective: To provide a management system, including policies and a framework to enable the effective management and implementation of all IT service.

3.1. 管理职责 management responsibility

管理层承担确保采用基于最佳实践的流程并得以维持的职责，是任何服务提供方达到ISO/IEC 20000-1要求的基础。

The role of management in ensuring best practice processes are adopted and sustained is fundamental for any service provider to meet the requirements of ISO/IEC 20000-1.

为确保实现承诺，应该指定一名高级管理人员，对服务管理计划负责。该高层管理者应该负责整体服务管理计划。

To ensure commitment an owner at senior level should be identified as being responsible for service management plans. This senior responsible owner should be accountable for the overall delivery of the service management plan.

高级管理者的职责应该包括为任何持续的或基于项目的服务改进活动安排资源。

The senior responsible owner's role should encompass resourcing for any continual or project based service improvement activities.

高级管理者应该得到有足够权限的决策层的支持，以定义服务方针并推行相关决策。

The senior responsible owner should be supported by a decision-taking group with sufficient authority to define policy and to enforce its decisions.

3.2. 文件要求 Documentation requirements

高级管理者应该确保提供服务管理方针、计划、程序和任何相关活动的审核证据。

The senior responsible owner should ensure that evidence is available for an audit of service management policies, plans and procedures, and any activities related to these.

服务管理计划和运行的很多证据应以文件的形式存在，文件可以是适合于目的的任何类型、格式或介质。

Much of the evidence of service management planning and operations should exist in the form of documents, which may be any type, form or medium suitable for their purpose.

下列文件通常被认为是服务管理计

The following documents are normally considered suitable

划的证据。

as evidence of service management planning.

a) 方针和计划；

a) policies and plans;

b) 服务文件；

b) service documentation;

c) 程序；

c) procedures;

d) 过程；

d) processes;

e) 过程控制记录；

e) process control records.

应该建立文件建立和管理的过程，以帮助保证满足所描述的特性。

There should be a process for the creation and management of documents to help ensure that the characteristics described are met.

应保护文件免受损坏，如恶劣环境或计算机灾难等。

Documentation should be protected from damage due, for example, to poor environmental conditions and computer disasters.

3.3. 能力、意识和培训 Competence, awareness and training

3.3.1. 总则 General

履行服务管理角色的人员应该具有基于适当教育、培训、技能和经验的相应能力。

Personnel performing work within service management should be competent on the basis of appropriate education, training, skills, and experience.

服务提供方应该：

The service provider should:

a) 确定服务管理中每个角色所需的能力；

a) determine the necessary competence for each role in service management;

b) 确保员工在广泛的业务范围内

b) ensure that personnel are aware of the relevance and

意义到他们活动的相关性和重要性，以及他们如何为达到质量目标做出贡献；

importance of their activities within the wider business context and how they contribute to the achievement of quality objectives;

c) 维护适当的教育、培训、技能和经验的记录；

c) maintain appropriate records of education, training, skills and experience;

d) 提供培训或采取其他措施以满足这些要求；

d) provide training or take other action to satisfy these needs;

e) 评估所采取措施的有效性。

e) evaluate the effectiveness of the actions taken.

3.3.2. 职业发展 Professional development

服务提供方应该开发并提高员工的专业技能。为实现上述目标，服务提供方应该采取以下措施：

The service provider should develop and enhance the professional competence of their workforce. Among the measures taken to achieve this, the service provider should address the following:

a) 招聘：客观检查职位申请人的详细信息（包括专业资格）的有效性，对照岗位说明书、服务管理目标和整体服务质量目标，识别申请人优点、缺点和潜力；

a) recruitment: with the objective of checking the validity of job applicants' details (including their professional qualifications) and identifying applicants' strengths, weaknesses and potential capabilities, against a job description/profile, service management targets and overall service quality objectives;

b) 计划：根据新的或扩展的服务（包括服务缩减）安置员工，安排服务管理人员的发展；制定后续计划和补充员工预期离职的空缺；

b) planning: with the objective of staffing of new or expanded services (also contracting services), using new technology, assigning service management staff to development project teams, succession planning and filling other gaps due to anticipated staff turnover;

c) 培训和开发：根据已识别的培训和发展要求的目标，制订培训和发展计划，并及时和有效的实施；

员工应该在相关的服务管理领域接受培训（如通过培训课程、自学、导师指导或在工作中学习），还应该培养员工的团队合作和领导能力。应该维护按时间进行的培训记录，包括培训内容的描述。

c) training and development: with the objective of identifying training and development requirements as a training and development plan and providing for timely and effective delivery.

Staff should be trained in the relevant aspects of service management (e.g. via training courses, self study, mentoring and on the job training) and their team-working and leadership skills should be developed. A chronological training record should be maintained for each individual, together with descriptions of the training provided.

3.3.3. 考虑的方法 Approaches to be considered

为了使用团队成员具备相应级别的能力，服务提供商应该确定适当比例的固定成员和临时成员的组成。服务提供商也应该确定具备技能的新成员和经过再培训的老成员的适当人员组成比例。

注：当支持员工的技能和数量发生重大变化，以及服务提供商在制定怎样提供服务的计划时，长期和临时雇员的平衡是相当重要的。

当建立最佳团队组合时，应该考虑以下因素：

a) 新增、变更的能力要求的短期或

In order to achieve teams of staff with appropriate levels of competence the service provider should decide on the optimum mix of short term and permanent recruits. The service provider should also decide on the optimum mix of new staff with the skills required and re-training of existing staff.

NOTE The optimum balance of short term and permanent recruits is particularly important when the service provider is planning how to provide a service during and after major changes to the number and skills of the support staff.

Factors that should be considered when establishing the most suitable combination of approaches include:

a) short or long term nature of new or changed

长期特点;	competencies;
b) 技巧和能力变更的频度;	b) rate of change in the skills and competencies;
c) 基于服务管理和改进计划, 预计的最大工作量、最小工作量以及所需要的技能;	c) expected peaks and troughs in the workload and skills mix required, based on service management and service improvement planning;
d) 具有合适能力的员工的可用性;	d) availability of suitably competent staff;
e) 人员流动率;	e) staff turnover rates;
f) 培训计划。	f) training plans.
针对所有员工, 服务提供方应该进行至少每年一次的员工绩效评审, 并采取相应措施。	For all staff, the service provider should review each individual's performance at least annually and take appropriate action.

4. 计划和实施服务管理 PLANNING AND IMPLEMENTING SERVICE MANAGEMENT

4.1. 服务管理的计划 (计划) Plan service management (Plan)

目标: 计划服务管理的实施和交付。	Objective: To plan the implementation and delivery of service management.
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4.1.1. 服务管理范围 Scope of service management

应在服务管理计划中定义服务管理范围。	The scope of service management should be defined as part of the service management plan.
例如, 可以定义以下内容:	For example, it may be defined by

a) 组织;

a) organization;

b) 位置;

b) location;

c) 服务;

c) service.

定义服务范围是管理职责的一部分（也是服务管理计划的一部分）。应该依据ISO/IEC 20000-1检查范围的适当性。

Management should define the scope as part of their management responsibilities (and as part of the service management plan). The scope should then be checked for suitability under ISO/IEC 20000-1.

注：运行变更的计划在9.2中描述。

NOTE Planning for operational changes is described in 9.2.

4.1.2. 计划的方法 Planning approaches

可用多个服务管理计划替代一个大的计划或方案。此时，基本的服务管理过程应该相互一致。同时也应该能够证实每个计划需求得到相应角色、职责和程序的管理。

Multiple service management plans may be used in place of one large plan or programme. Where this is the case the underlying service management processes should be consistent with each other. It should also be possible to demonstrate how each planning requirement is managed by it to the corresponding roles, responsibilities and procedures.

服务管理计划应该是将客户需求和高层管理意图转化为服务的过程的一部分，为指导发展提供路线。

Service management planning should form part of the process for translating customers' requirements and senior management intentions into services, and for providing a route map for directing progress.

服务管理计划应该包括：

A service management plan should encompass:

a) 服务管理（或部分服务管理）的实施；

a) implementation of service management (or part of service management);

- | | |
|--------------------------------|---|
| b) 服务管理过程的交付; | b) delivery of service management processes; |
| c) 服务管理过程的变更; | c) changes to service management processes; |
| d) 服务管理过程的改进; | d) improvements to service management processes; |
| e) 新建服务（针对影响服务管理商定范围内过程的新建服务）; | e) new services (to the extent that they affect processes within the agreed scope of service management). |

4.1.3. 需要考虑的事态 Events to be considered

服务管理计划应符合服务管理过程和由事态触发的服务变更，如：

The service management plan should cater for service management process and service changes triggered by events such as:

- | | |
|-------------------|---|
| a) 服务改进; | a) service improvement; |
| b) 服务变更; | b) service changes; |
| c) 基础设施的标准化; | c) infrastructure standardization; |
| d) 法律的变化; | d) changes to legislation; |
| e) 法规的变化，如本地税率改变; | e) regulatory changes, e.g. local tax rate changes; |
| f) 行业规范或行业规范的违背; | f) deregulation or regulation of industries; |
| g) 合并和并购。 | g) mergers and acquisitions. |

4.1.4. 计划的范围和内容 Scope and contents of the plan

服务管理计划应该定义：

A service management plan should define:

- | | |
|-------------------|--|
| a) 服务提供方的服务管理范围; | a) the scope of the service provider's service management; |
| b) 服务管理要实现的目标和要求; | b) the objectives and requirements that are to be achieved |

	by service management;
c) 实现定义目标所需的资源、设施和预算;	c) the resources, facilities and budgets necessary to achieve the defined objectives;
d) 管理角色和职责架构, 包括高层管理者代表, 过程所有人和供应商的管理;	d) the framework of management roles and responsibilities, including the senior responsible owner, process owners and management of suppliers;
e) 服务管理过程之间的接口, 以及活动和/或过程之间的协调方式;	e) the interfaces between service management processes and the manner in which the activities and/or processes are to be co-ordinated;
f) 用于达到预定目标的识别、评估和管理问题及风险的方法;	f) the approach to be taken in identifying, assessing and managing issues and risks to the achievement of the defined objectives;
g) 按日期描述的资源安排, 包括可用的资金、技能和资源;	g) a resource schedule expressed in terms of the dates on which funds, skills, and resources should be available;
h) 改变计划中已定义的计划和服務的方法;	h) the approach to changing the plan and the service defined by the plan;
i) 服务提供方如何证明持续的质量控制(如期中的审核);	i) how the service provider will demonstrate continuing quality control (e.g. interim audits);
j) 将被执行的过程;	j) the processes that are to be executed;
k) 适当的支持过程的工具;	k) tools as appropriate to support the processes.

4.2. 实施服务管理和提供服务 (实施) Implement service management and provide the services (Do)

目标：实施服务管理的目标和计划。	Objective: To implement the service management objectives and plan.
如果服务不能达到实施ISO/IEC 20000-1所描述的要求，就不可能获得达到ISO/IEC 20000所要求的基于最佳实践的服务管理过程的能力。	Attainment of best practice service management processes capable of meeting the requirements of ISO/IEC 20000 will not be achieved if the original services do not meet the requirements outlined for the implementation in ISO/IEC 20000-1.
一旦实施，应该对服务和 service 管理过程进行维护。	Once implemented the service and service management processes should be maintained.
应该按照条款4.3实施评审。	Reviews should take place in accordance with 4.3.
注：适合于计划和初始实施的人员，可能不适合于运作。	NOTE The person that is appropriate for the planning and initial implementation may not be suitable for the ongoing operation.
4.3. 监视、测量和评审（检查） Monitoring, measuring and reviewing (Check)	
目标：对服务管理目标和计划的达成状况进行监视、测量和评审。	Objective: To monitor, measure and review that the service management objectives and plan are being achieved.
服务提供方应该计划和实施对服务、服务管理过程及辅助系统的监视、测量、分析和评审。应该进行监视、测量和评审的内容有：	The service provider should plan and implement the monitoring, measurement, analysis and review of the service, the service management processes and associated systems. Items that should be monitored, measured, and reviewed include:
a) 既定服务目标的达成；	a) achievement against defined service targets;

b) 客户满意度;

b) customer satisfaction;

c) 资源的利用;

c) resource utilisation;

d) 趋势;

d) trends;

e) 严重不符合;

e) major non-conformities.

分析的结果应该作为服务改进计划的输入。

The results of the analysis should provide input to a plan for improving the service.

如同测量和分析等服务管理活动, 高层管理者可能还需要采用内审或其他方式进行检查。在确定内审和检查的频次时, 应该考虑过程中的风险程度、运作的频率及过去发生的问题。应该对内审和检查进行计划, 由具备相应能力的人员实施并记录。

As well as service management activities on measurement and analysis senior management may need to make use of internal audits and other checks. When deciding the frequency of such internal audits and checks, the degree of risk involved in a process, its frequency of operation and its past history of problems are among the factors that should be taken into account. Internal audits and checks should be planned, carried out competently and recorded.

4.4. 持续改进 (改进) Continual improvement (Act)

目标: 提高服务交付和管理的效率和效力。

Objective: To improve the effectiveness and efficiency of service delivery and management.

4.4.1. 方针 Policy

服务提供方应该意识到总是存在使服务交付更有效力和效率的机会。应该发布服务质量和改进的方针。

Service providers should recognize that there is always the potential to make delivery of services more effective and efficient. There should be a published policy on service quality and improvement.

所有参与服务管理和改进的人员应该意识到服务质量方针和自身根据方针来实现目标的贡献。

All those involved in service management and service improvement should be aware of the service quality policy and their personal contribution to the achievement of the objectives laid out within this policy.

特别是，服务提供商参与服务管理的所有员工，应该详细的理解服务管理过程的含义。

In particular all the service provider's staff involved in service management should have a detailed understanding of the implications of this on service management processes.

在影响服务质量和客户要求的问题上，应该建立服务提供方自身管理架构、客户以及服务提供方的供应商之间的有效联络渠道。

There should be effective liaison within the service provider's own management structure, customers and the service provider's suppliers on matters affecting service quality and customer requirements.

4.4.2. 服务改进计划 Planning for service improvements

服务提供方应该根据自身要求和客户期望，采用系统的、协作的方法，进行服务改进，以满足方针要求。

Service providers should adopt a methodical and coordinated approach to service improvement to meet the requirements of the policy, from their own and from their customer's perspective.

在实施改进服务计划之前，应该记录服务质量和级别，以作为实际改进比较的基准。实际的改进应与以前的改进相比较，以评估变更的有效性。

Before implementing a plan for improving the service, service quality and levels should be recorded as a baseline against which the actual improvements can be compared. The actual improvement should be compared to the predicted improvement to assess the effectiveness of the change.

注1：服务改进要求可能来源于所有

NOTE 1 Service improvement requirements can come from

流程。	all processes.
服务供应商应鼓励他们的员工和客户提供改进服务的建议。	Service providers should encourage their staff and customers to suggest ways of improving services.
注2 服务改进可以通过建议方案、质量环、用户组和联络会议进行实施。	NOTE 2 This may be done using suggestion schemes, quality circles, user groups and liaison meetings.
服务改进目标应该可测量，与业务目标关联，并文件化到在计划。	Service improvement targets should be measurable, linked to business objectives and documented in a plan.
应该对服务改进进行主动的管理，并按正式商定的目标进行监视。	Service improvement should be actively managed and progress should be monitored against formally agreed objectives.

5. 计划和实施新服务或变更的服务 PLANNING AND IMPLEMENTING NEW OR CHANGED SERVICES

目标：确保在商定的成本和服务质量条件下，管理和交付新建和变更的服务。	Objective: To ensure that new services and changes to services will be deliverable and manageable at the agreed cost and service quality.
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5.1. 考虑的主题 Topics for consideration

新建或变更的服务的相关计划应该包括内容的评审：	Planning for new or changed services should include reviewing:
a) 预算；	a) budgets;
b) 人力资源；	b) staff resources;

- | | |
|----------------------------------|--|
| c) 现有的服务等级; | c) existing service levels, |
| d) SLA、其他目标或服务承诺; | d) SLAs and other targets or service commitments; |
| e) 现有的服务管理流程、程序和文件; | e) existing service management processes, procedures and documentation; |
| f) 服务管理范围, 包括以前从范围中排除的服务管理流程的实施。 | f) the scope of service management, including the implementation of service management processes previously excluded from the scope. |

5.2. 变更记录 Change records

所有服务的变更应该在变更记录中反映。

All service changes should be reflected in Change Management records.

包括下列事项的计划:

This includes plans for:

- | | |
|---------------|---|
| a) 员工招聘/再培训; | a) staff recruitment/retraining; |
| b) 再分配; | b) relocation; |
| c) 用户培训; | c) user training; |
| d) 对变更的沟通; | d) communications about the changes; |
| e) 技术支持特性的变更; | e) changes to the nature of the technology supported; |
| f) 服务的关闭。 | f) formal closure of services. |

6. 服务交付过程 SERVICE DELIVERY PROCESSES

6.1. 服务等级管理 Service level management

目标：定义、协商、记录和管理服务等级。

Objective: To define, agree, record, and manage levels of service.

6.1.1. 服务目录 Service catalogue

服务目录应该定义所有服务。服务目录可以参考SLA，并应该用作评审SLA自身不确定性的材料。

A service catalogue should define all services. It can be referenced from the SLA and should be used to hold material considered volatile for the SLA itself.

应该对服务目录进行维护，并保持更新。

The service catalogue should be maintained and kept up-to-date.

注：服务目录应该包含以下通用信息：

NOTE The service catalogue can include generic information such as:

a) 服务名称；

a) the name of the service;

b) 目标，如响应和安装打印机的时间、重大故障后恢复服务的时间；

b) targets, e.g. time to respond or install a printer, time to re-instate a service after a major failure;

c) 联系方式；

c) contact points;

d) 服务时间和例外；

d) service hours and exceptions;

e) 安全处理。

e) security arrangements.

服务目录是设置客户期望的关键文件，应该被客户和支持员工很容易访问和获取。

The service catalogue is a key document for setting customer expectation and should be easily accessible and widely available to both customers and support staff.

6.1.2. 服务级别协议（SLAs） Service level agreements (SLAs)

服务应该在服务等级协议（SLA）中正式文件化。SLA也应由客户高层代表和服务提供方代表认可。SLA如同其描述的服务一样，遵循变更管理。

A service should be formally documented in a service level agreement (SLA). The SLA should be formally authorized by senior customer and service provider representatives. The SLA should be subject to change management, as is the service that it describes.

客户的业务需求和预算，应该作为SLA的内容、结构和目标的强制内容。应该从客户的角度，来定义所交付服务的测量目标。

The customer's business needs and budget should be the defining force for the content, structure and targets of the SLA. The targets, against which the delivered service should be measured, should be defined from a customer perspective.

SLA应该仅关注目标中与服务中最重要方面相关的适当部分。

The SLAs should include only an appropriate subset of the targets to focus attention on the most important aspects of the service.

注1：目标太多可能导致混乱，导致经费超支。

NOTE 1 Too many targets can create confusion and lead to excessive overheads.

SLA中应该包含最基本内容或可以直接从SLA中引用的内容有：

The minimum content that should be in an SLA or that can be directly referenced from an SLA is:

- a) 服务的简述；
- b) 有效期和/或SLA变更控制机制；
- c) 授权细节；
- d) 沟通方式的简述，包括报告；

- a) brief service description;
- b) validity period and/or SLA change control mechanism;
- c) authorization details;
- d) brief description of communications, including reporting;

e) 授权参与应急处理、事件和问题纠正、恢复或临时措施的人员联系信息；	e) contact details of people authorized to act in emergencies, to participate in incidents and problem correction, recovery or workaround;
f) 服务时间，如从9: 00到17: 00，例外的日期（如周末、公共假日），重要业务时间及非工作时间的安排；	f) service hours, e.g. 09:00 h to 17:00 h, date exceptions (e.g. weekends, public holidays), critical business periods and out of hours cover;
g) 计划和商定的服务中断，包括需要发出的通知和每阶段的次数；	g) scheduled and agreed interruptions, including notice to be given, number per period;
h) 客户的职责，如安全；	h) customer responsibilities, e.g. security;
i) 服务提供方职责和义务，如安全；	i) service provider liability and obligations e.g. security;
j) 影响和优先级指南；	j) impact and priority guidelines;
k) 升级和通知过程；	k) escalation and notification process;
l) 投诉程序；	l) complaints procedure;
m) 服务目标；	m) service targets;
n) 工作量限制（最大及最小工作量），如支持商定用户数量/工作量、系统吞吐量等服务能力；	n) workload limits (upper and lower), e.g. the ability of the service to support the agreed number of users/volume of work, system throughput;
o) 基本财务管理信息，如付费帐号等；	o) high level financial management details, e.g. charge codes etc;

- | | |
|--------------------|---|
| p) 服务中断时采取的措施； | p) action to be taken in the event of a service interruption; |
| q) 后勤管理过程 | q) housekeeping procedures; |
| r) 术语表； | r) glossary of terms; |
| s) 支持和相关服务； | s) supporting and related services; |
| t) 在SLA中规定条款的例外情况。 | t) any exceptions to the terms given in the SLA. |

注2：如果所被引用文件受变更管理流程控制，一些不确定信息或多个SLA的通用信息（如具体的联系信息）可以直接从SLA引用，而不影响SLM过程的质量。

NOTE 2 Volatile information, or information common to many SLAs (such as contact details) can be referenced from the SLA without impacting the quality of SLM processes as long as the referenced documents are also under the control of the change management process.

注3：连续性计划和核算/预算细节通常从SLA中引用。

NOTE 3 Continuity plan and details of accounting & budgeting are normally referenced from the SLA.

注4：属于表通常单独集中保存，并适用于所有文件，包括服务目录。

NOTE 4 A glossary of terms is normally held in one place and is common to all documents, including the service catalogue.

6.1.3. 服务等级管理过程 (SLM) Service level management (SLM) process

重要的业务变更，如增长、业务重组、合并、以及客户需求的变更，都可能需要调整、重定义服务等级，甚至临时暂停。SLM过程应灵活的适应这些变更。SLM流程应确保服务提供方在计划、实施和持续的服务交付管理中	Major business changes, due, for example, to growth, business reorganizations and mergers, and changing customer requirements, can require service levels to be adjusted, redefined or even temporarily suspended. The SLM process should be flexible to accommodate these changes. The SLM process should ensure that the service
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保持对客户要求的关注。

provider remains focused on the customer throughout the planning, implementation, and ongoing management of service delivery.

服务提供方应该获得足够的信息，以确保理解客户的业务驱动和要求。

The service provider should be given adequate information to enable them to understand their customer's business drivers and requirements.

SLM过程应该管理和协调服务等级的保障因素，包括：

The SLM process should manage and coordinate contributors of the service levels, to include:

a) 服务要求和预期服务工作量特征的协议；

a) agreement of the service requirements and expected service workload characteristics;

b) 服务目标的协议；

b) agreement of service targets;

c) 服务等级实现、工作量的测量和报告，以及服务目标不能完成的解释(见6.2)；

c) measurement and reporting of the service levels achieved, workloads and an explanation if the agreed targets are not met (see 6.2);

d) 纠正措施的发起；

d) initiation of corrective action;

e) 改进服务计划的输入；

e) input to a plan for improving the service.

SLM过程应该鼓励服务供应商和客户采取积极态度以确保其对服务有连带责任。

The process should encourage both the service provider and the customer to develop a proactive attitude ensuring that they have joint responsibility for the service.

客户满意度是服务级别管理中的重要部分，应该通过客观的测量进行验证，同时SLA中的服务目标也应该进行客观的测量。SLM过程应该与业务

Customer satisfaction is an important part of service level management but it should be recognized as being a subjective measurement, whereas service targets within an SLA should be objective measurements. The SLM process

关系和供应商过程紧密相关。

should work closely with the business relationship and supplier management processes.

6.1.4. 服务协议的支持 Supporting service agreements

交付服务所依赖的支持服务应该形成文件，并与相关供应商达成一致。这包括内部小组提供作为服务提供方提供的部分服务。

The supporting services on which the delivered service depends should be documented and agreed with each supplier. This includes internal groups providing part of the service provider's service.

6.2. 服务报告 Service reporting

目标：为做出基于知情的决策和有效沟通，编制商定的、及时的、可信赖的、准确的报告。

Objective: To produce agreed timely, reliable, accurate reports for informed decision making and effective communication.

注：所有服务管理流程的成功依赖于服务报告中提供信息的利用。

NOTE The success of all service management processes is dependent on the use of the information provided in service reports.

6.2.1. 方针 Policy

服务报告的要求应该与客户和内部管理层协商，并记录。

The requirements for service reporting should be agreed and recorded for customers and internal management.

服务监控和报告包括服务的所有可测量的方面，并提供当前和历史分析。

Service monitoring and reporting encompasses all measurable aspects of the service, providing both current and historical analysis.

如果存在多个供应商，主供应商和分包商的报告应该反应主供应商和分包供应商之间的关系。例如，主供应

Where there are multiple suppliers, lead suppliers and sub-contracted suppliers the reports should reflect the relationships between suppliers. For example, a lead

商应报告他们所提供的全部服务，包括其作为客户服务的一部分进行管理的、由分包供应商提供的服务。	supplier should report on the whole of the service they provide, including any services by sub-contracted suppliers that they manage as part of the customer's service.
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6.2.2. 服务报告中的目标和质量检查 Purpose and quality checks on service reports

服务报告应及时、清晰、可靠和简明。	Service reports should be timely, clear, reliable, and concise.
报告应该适合于接受方的需求，并充分准确以可作为决策支持工具。	They should be appropriate to the recipient's needs and of sufficient accuracy to be used as a decision support tool.
描述方法应有助于容易理解报告，如使用图表。	The presentation should aid the understanding of the reports so that they are easy to assimilate, e.g. use of charts.
应该形成以下几种类型报告：	Several types of report should be produced:
a) 描述所发生事件的响应报告；	a) reactive reports which show what has happened;
b) 主动报告，对重要事件提前警告，使得可以预先采取预防措施（如，报告即将发生的违背SLA的事件）；	b) proactive reports, which give advance warning of significant events, thereby enabling preventive action to be taken beforehand (for example reports of impending breaches in SLAs);
c) 描述计划活动的预先计划报告。	c) forward scheduled reports showing planned activities.

6.2.3. 服务报告 Service reports

服务提供方向客户和管理层提交的报告，应该包括以下内容：	The service provider should produce reports for customers and management covering:
a) 对照服务等级目标的绩效，如损	a) performance against service level targets, e.g. outage

失报告、成绩；	reports, achievements;
b) 与标准的不符合项；	b) non-compliance with standards;
c) 工作量的特征和数量信息，如事件、问题、变更、任务、分类、位置、客户、季节趋势、优先次序，请求帮助数量等；	c) workload characteristics and volume information, e.g. incidents, problems, changes and tasks, classification, location, customer, seasonal trends, mix of priorities, numbers of requests for help;
d) 重大事件的绩效报告，如变更、版本发布；	d) performance reporting following major events, e.g. change, and releases;
e) 阶段性的趋势信息（如，每天、每周、每月）；	e) trend information by period (e.g. day, week, month, period);
f) 报告每个过程的信息，如事件数量和最被常问及的问题、基础设施的不可靠组件，资源/成本集中的任务；	f) reports that include information from each process, e.g. the number of incidents and the most frequently asked questions, unreliable components of the infrastructure, resource/cost intensive tasks;
g) 远景重要事项的报告和计划的工作量。	g) reports to highlight future and scheduled workloads.

6.3. 服务持续性和可用性管理 Service continuity and availability management

目标：确保在所有条件下达到承诺给客户的服务连续性和可用性。

Objective: To ensure that agreed service continuity and availability commitments to customers can be met in all circumstances.

注：重大服务失效或灾难可能由多种原因引起，包括拒绝服务、攻击、重大的病毒破坏、未经授权的访问或

NOTE Major service failures or disasters can occur for many reasons including denial of service, attack, major virus outbreak, access to the premises not allowed or a

自然灾害。

natural disaster.

6.3.1. 总则 General

应该基于客户业务优先级、服务等级协议和评估的风险来识别，服务连续性和可用性要求。服务提供方应该具备足够的服务能力和可行的工作计划，确保从普通到重大服务失效的任何环境下都能达到商定的要求。服务提供方应该对已知数据或用户量的增加/减少、预计工作量的高峰和低谷，或其他已知的未来变更进行计划。

Service continuity and availability requirements should be identified on the basis of the customers' business priorities, service level agreements and assessed risks. The service provider should maintain sufficient service capability together with workable plans designed to ensure that agreed requirements can be met in all circumstances from normal through to a major loss of service. The service provider should plan for known data or user volume increases or decreases, expected peaks and troughs in workload and any other known future changes.

需求应该包括访问权限和响应时间，以及端到端的系统组件的可用性。

Requirements should include access rights and response times as well as end-to-end availability of system components.

应该综合维持服务可用性和服务持续管理，以确保维护协议服务级别目标的达成。这些要求将对为满足服务的可用性，而进行的活动、努力和资源分配有重要影响。

Service availability and service continuity management should work together with the aim of ensuring that agreed service levels are maintained. These requirements should have a major influence on the actions, efforts and resources allocated to matching the availability of services that support them.

确保维护必要可用性的过程，应该包括受客户和其他服务提供商控制的服务交付的组件。

Processes to ensure that required availability is maintained should include those elements of the service delivery that are under the control of the customer or other service providers.

6.3.2. 可用性的监视和活动 Availability monitoring and activities

可用性管理应该包括：

- a) 监视和记录服务的可用性；
- b) 维护精确的历史数据；
- c) 对照SLA所定义的要求，以识别可用性目标的不符合事项；
- d) 文件化不符合项，并进行评审；
- e) 预计未来的可用性；
- f) 如有可能，应该预计潜在问题，并采取预防措施。

应该确保服务所有组件的可用性，记录所采取的纠正措施并实施。

Availability management should:

- a) monitor and record availability of the service
- b) maintain accurate historical data;
- c) make comparisons with requirements defined in SLAs to identify non-conformance to the agreed availability targets;
- d) document and review non-conformance;
- e) predict future availability;
- f) where possible, potential issues should be predicted and preventive action taken.

It should ensure availability of all components of the service, with corrective actions recorded and acted upon.

6.3.3. 服务持续性策略 Service continuity strategy

服务提供方应该开发和维护策略，以定义达到服务持续性义务的总体方法。这包括风险评估、对商定的服务时间和关键业务周期的考虑。服务提供方应该与每个客户群和服务进行协商：

- a) 服务丧失的最长可接受持续期限；

The service provider should develop and maintain a strategy that defines the general approach to be taken to meeting service continuity obligations. This should include risk assessment and take into account agreed service hours and critical business periods. The service provider should agree for each customer group and service:

- a) maximum acceptable continuous period of lost service;

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| b) 降级服务的最长可接受期限； | b) maximum acceptable periods of degraded service; |
| c) 在服务恢复期间，可接受的降级服务等级。 | c) acceptable degraded service levels during a period of service recovery. |

应该按商定的时间间隔评审业务连续性策略，但至少一年一次。

The continuity strategy should be reviewed at agreed intervals, at least annually.

应该正式商定策略的任何变更。

Any changes to the strategy should be formally agreed.

6.3.4. 服务持续性计划和测试 Service continuity planning and testing

服务提供方应该确保：

The service provider should ensure that:

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| a) 对服务和系统组件的关系的持续性计划的考虑； | a) continuity plans take into account dependencies between service and system components; |
| b) 记录并维护支持服务连续性的服务连续性计划和其他文件； | b) service continuity plans and other documents required to support service continuity are recorded and maintained; |
| c) 应该清晰的分配调用持续计划的职责，并清晰的计划对每个目标采取措施的职责； | c) responsibility for invoking continuity plans is clearly assigned, and plans clearly allocate responsibility for taking action against each objective; |
| d) 在重大服务失效或灾难时，服务恢复所需的备份数据、文件、软件、任何设备和人员，应该快速可用； | d) backups of data, documents and software, and any equipment and staff necessary for service restoration are quickly available following a major service failure or disaster; |
| e) 至少应该在远程的安全地点存储和维护所有服务持续文件的 | e) at least one copy of all service continuity documents should be stored and maintained at a secure remote |

一份拷贝，同时该处具有使得文件可用的必要设备；	location, together with any equipment that is necessary to enable its use;
f) 员工应该理解在启动/执行计划中的职责，并能访问服务连续性文件。	f) staff understand their role in invoking and/or executing the plans; and are able to access service continuity documents.
服务持续性计划和相关文件（如合同）应该与变更管理流程和合同管理流程相关联。	Service continuity plans and related documents (e.g. contracts) should be linked to the change management process and the contract management process.
在系统和服务变更被批准之前，以及与客户签订新的或修订的客户要求之前，应该评估对服务持续性计划和相关文件（如合同）的影响。	Service continuity plans and related documents (e.g. contracts) should be assessed for impact prior to system and service changes being approved, and prior to significant new or amended customer requirements being agreed.
应经常进行服务持续性计划的测试，以确保服务连续性计划有效，并当面临系统、流程、人员和业务需求变更时，仍然保持有效。测试应该在商定的系列目标基础上，由客户和服务提供方共同进行。应该对测试的失败进行记录和评审，并输入服务改进计划。	Testing should be undertaken at a frequency sufficient to gain assurance that continuity plans are effective, and remain so in the face of changing systems, processes, personnel and business needs. Testing should be a joint involvement between customer and service provider based upon an agreed set of objectives. Test failures should be documented and reviewed to input to a plan for improving the service.

6.4. IT 服务预算和核算 Budgeting and accounting for IT services

目标：提供服务成本的预算和核算。	Objective: To budget and account for the cost of service provision.
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6.4.1. 总则 General

本部分包括IT服务预算和财务核算。实际上许多组织都涉及服务收费问题。但是收费是一个非强制性的活动，因此本标准并没有涉及。建议服务提供商在实施收费时，应明确收费机制，并让相关各方了解。

This section covers budgeting and accounting for IT services. In practice, many service providers will be involved in charging for such services. However, since charging is an optional activity, it is not covered by the standard. Service providers are recommended that where charging is in use, the mechanism for doing so is fully defined and understood by all parties.

许多财务决策已经超出了服务管理的范畴，应更该多长时间、以什么形式、提供什么财务信息往往取决于外部的要求。本部分只关注为达到标准要求而采取的措施。但还应该考虑更多的要求，因为这些要求会对一些方针和程序带来影响。所有的财务措施都应该与整个服务提供方组织的财务管理相一致。

Responsibility for many of the financial decisions will lie outside the sphere of the service management arena and the requirements for what financial information is to be provided, in what form and at what frequencies may be dictated from outside. The provisions of this section are focused on the practices that should be followed to satisfy the requirements of the standard. However, wider requirements should also be taken into account as they will impact on some of the policies and procedures defined. All accounting practices used should be aligned to the wider accountancy practices of the whole of the service provider's organization.

6.4.2. 策略 Policy

应建立服务的财务管理方针。方针应定义预算编制和财务核算的目标。

There should be a policy on the financial management of services. The policy should define the objectives to be met by budgeting and accounting.

方针还应该定义需要执行哪些预算和核算活动的细节，并考虑以下因素：

The policy should also define the detail to which budgeting and accounting are performed, taking into consideration the:

a) 需要核算的成本类型；

b) 经费的分摊，例如统一费率、固定比例或基于可变要素的规模等；

c) 客户业务的细化程度，如业务单元作为一个单元，进一步细分到或按区域；

d) 管理预算差异的规则，如多大的差异将提交给高层管理者；

e) 与服务等级管理相关联。

在预算和核算中的投入等级应该根据客户要求及服务提供方和供应商在方针中所定义的财务方面的要求。

注：在商业环境下，服务提供方可能需要为财务管理投入更多的时间及精力。相反，对只需简单识别成本的服务提供方，其相应的财务管理可能会比较简单。

无论在财务管理上的其它方针，所有组织都应该履行预算和核算。

a) cost types to be accounted for;

b) apportionment of overhead costs, e.g. flat rate, fixed percentage, or based on the size of the variable element;

c) granularity of the customer's business, e.g. business unit as one unit, subdivided into department, or by locations;

d) rules governing the handling of variances against budgets, e.g. size of variance that will be escalated to senior management;

e) links to service level management.

The level of investment in budgeting and accounting processes should be based on the needs of the customers, service provider and suppliers for financial detail as defined in the policy.

NOTE Service providers operating in a commercial environment might need to invest considerably more time and effort in their financial management. Conversely, for service providers where simple identification of costs is sufficient the financial management can be much simpler.

Budgeting and accounting should be performed by all service providers, whatever their other policies on financial management.

6.4.3. 预算 Budgeting

预算编制应该考虑在预算期内计划的服务变更、预算对额外可用资金的要求、资金短缺的管理计划。

Budgeting should take into account the planned changes to services during the budget period and where budgetary requirements exceed available funds, plan for the management of shortfalls.

预算编制应该考虑季节变化，以及短期计划内的服务成本和收费的变更。

Budgeting may take into account factors such as seasonal variations and short term planned changes to service costs and charges.

对照预算的成本跟踪，应该为预算差异提供预警信息。

Cost tracking against the budget should provide early warning of variances against budgets.

应该建立管理造成预算差异因素的过程。

There should be a process that manages the implications of variances against budget.

预算管理和成本跟踪应该为服务的运营和变更提供支持，确保在整个财务年度内保持相应的服务等级。

Budgeting and cost tracking should support planning to operate and change the services so that service levels can be maintained throughout the year.

6.4.4. 财务 Accounting

应该在商定的阶段内，根据商定的等级，利用核算过程跟踪成本。

Accounting processes should be used to track costs to an agreed level of detail over an agreed period of time.

应该根据成本效益的比较，进行有关服务提供的决策。

Decisions about service provision should be based on cost effectiveness comparisons.

成本模型应说明服务提供的成本。

Cost models should be able to demonstrate the costs of service provision.

核算应该说明费用的超支、节余和补救的费用等，并让读者理解地等级服务或服务丧失所带来的成本。	Accounts should demonstrate over and under-spending/recovery; and should allow the reader to understand the costs of low service levels or loss of service.
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6.5. 容量管理 Capacity management

目标：确保服务提供方一直保持有效的容量去满足当前和未来客户业务需求。	Objective: To ensure that the service provider has, at all times, sufficient capacity to meet the current and future agreed demands of customer's business needs.
明确需要为客户提供什么样的服务，将有助于更好地了解对服务当前及未来的业务要求。	The current and expected business requirements for services should be understood in terms of what the business will need to enable it to deliver to its customers.
应该将业务预期和工作量估计转换成具体的要求，并形成文件。	Business predictions and workload estimates should be translated into specific requirements and documented.
工作量和环境变化的结果应该是可预期的；应该适当的收集并分析组件和资源当前及历史的利用状况，以支持该过程。	The result of variations in workload or environment should be predictable; data on current and previous component and resource utilization at an appropriate level should be captured and analysed to support the process.
容量管理应该是所有性能和容量问题的关注点。	Capacity management should be the focal point for all performance and capacity issues.
能力管理过程应该通过服务的规模和模式化管理，给开发新的或变更的服务提供直接支持。	The process should provide direct support to the development of new and changed services by providing sizing and modelling of services.
应该根据服务变更频率、服务量、变更管理报告和客户业务等信息，按合	A capacity plan documenting the actual performance of the infrastructure and the expected requirements should be

适的次数文件化基础设施实际性能和预期要求的能力管理计划。

produced at a suitable frequency taking into account the rate of change in services and service volumes, information in the change management reports and customer business.

应该每年至少进行一次能力管理计划的编制。能力管理计划应该说明为达到业务要求需要花费成本的项目，并建议保证达到SLA中所定义的服务等级目标的解决方案。

It should be produced at least annually. This should document costed options for meeting the business requirements and recommend solutions to ensure achievement of the agreed service level targets as defined in the SLA.

应该清楚了解技术基础设施及其当前和预期的能力。

The technical infrastructure and its current and projected capabilities should be well understood.

6.6. 信息安全管理 Information security management

目标：有效管理所有服务活动的信息安全。

Objective: To manage information security effectively within all service activities.

6.6.1. 总则 General

信息安全是用于识别、控制、保护信息以及用于存储、传输、处理信息的设备的方针和程序的体系化结果。

Information security is the result of a system of policies and procedures designed to identify, control and protect information and any equipment used in connection with its storage, transmission and processing.

服务提供方承担信息安全专家角色的员工应熟悉标准《ISO/IEC 17799，信息技术-安全技术-信息安全管理实用规则》。

The service provider's staff with specialist information security roles should be conversant with ISO/IEC 17799, Information technology — Security techniques — Code of practice for information security management.

6.6.2. 识别和分类信息资产 Identifying and classifying information assets

服务提供商应该：

- a) 维护提供服务所需要的信息资产清单（例如计算机、通信设备、环境设备、文档和其他信息）；
- b) 根据其服务的重要性以及所要求的保护级别对信息资产进行分类，并指定一定所有者来负责安全保护；
- c) 信息资产负责人可能把日常安全管理职责委派给他人，但仍需承担信息资产保护的责任。

The service provider should:

- a) maintain an inventory of the information assets (for example, computers, communications, environmental equipment, documents and other information) that are necessary to deliver services;
- b) classify each asset according to its criticality to the service and the level of protection it requires, and nominate an owner to be accountable for providing that protection;
- c) accountability for asset protection should rest with the asset owner, although they may delegate day-to-day security management responsibilities.

6.6.3. 安全风险评估实践 Security risk assessment practices

安全风险评估应该：

- a) 按商定的时间间隔进行；
- b) 保留记录；
- c) 依据变化（业务需求、流程和配置的变化）进行维护；
- d) 帮助了解风险对管理服务的影响；
- e) 根据已实施的风险控制类型进行基于知情的决策。

Security risk assessment should:

- a) be performed at agreed intervals;
- b) be recorded;
- c) be maintained during changes (of changing business needs, processes and configurations);
- d) help understanding of what could impact a managed service;
- e) inform decisions regarding the types of controls to be operated.

6.6.4. 信息资产的风险 Risks to information assets

信息资产风险评估应该参考以下因素：

- a) 特性（例如软件漏洞、运行错误、通信失效）；
- b) 可能性；
- c) 潜在的业务影响；
- d) 历史经验；

Risks to information assets should be assessed by reference to:

- a) their nature (e.g. software malfunction, operating errors, communications failure);
- b) likelihood;
- c) potential business impact;
- d) past experience.

6.6.5. 信息的安全和可用性 Security and availability of information

在评估风险时应考虑以下因素：

- a) 敏感信息被泄漏给未经授权方；
- b) 不准确、不完整或无效（例如：欺诈）信息；
- c) 无法获取信息（例如，电力故障）；
- d) 提供服务所需设备的物理损坏或破坏。

In assessing risks, due regard should be paid to the following:

- a) disclosure of sensitive information to unauthorized parties;
- b) inaccurate, incomplete or invalid (e.g. fraudulent) information;
- c) information being unavailable for use (e.g. due to power failure);
- d) physical damage to, or destruction of equipment necessary to provide services.

还应考虑信息安全方针目标，满足客户特定安全的要求（例如可用性登

Account should also be taken of information security policy objectives, the need to meet customers' specified security

记) 以及适用的法律法规。

requirements (e.g. availability levels), and statutory or regulatory requirements that apply.

6.6.6. 控制措施 Controls

除了ISO/IEC 20000中包含的控制措施(如, 服务连续性)以外, 服务提供商依据良好的信息管理实践, 实施以下控制措施:

In addition to other controls that might be justified and advice contained elsewhere in this part of ISO/IEC 20000 (e.g. on service continuity), service providers should operate the following controls as a matter of good information security management practice:

a) 高层管理者应该定义信息安全方针, 达给职工和客户, 并确保有效的实施;

a) senior management should define their information security policy, communicate it to staff and customers, and act to ensure its effective implementation;

b) 应定义信息安全管理角色和职责, 并分配给相关责任人;

b) information security management roles and responsibilities should be defined, and allocated to post holders;

c) 管理者代表(可能由最高责任人担任本角色)应该监视和维护信息安全策略的有效性;

c) a representative management group (the role might be undertaken by the senior responsible owner) should monitor and maintain the effectiveness of the Information Security Policy;

d) 重要信息安全角色应参加信息安全培训;

d) staff with significant security roles should receive information security training;

e) 所有员工应了解信息安全方针;

e) all staff should be made aware of the information security policy;

f) 在风险评估和控制措施的实施中专家的帮助;

f) expert help on risk assessment and control implementation should be available;

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| g) 变更应该不能危及控制措施的有效运行； | g) changes should not compromise the effective operation of controls; |
| h) 应该根据事件管理过程报告信息安全事件，并启动响应措施。 | h) information security incidents should be reported in line with incident management procedures and a response initiated. |

6.6.7. 文件和记录 Documents and records

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| 应该定期分析记录，并提供以下信息给管理层： | Records should be analysed periodically to provide management with information on: |
| a) 信息安全方针的有效性； | a) effectiveness of information security policy; |
| b) 信息安全事件呈现的趋势； | b) emerging trends in information security incidents; |
| c) 服务改进计划的输入； | c) input to a plan for improving the service; |
| d) 信息、资产和系统的访问控制； | d) control over access to information, assets, and systems. |
| 信息安全管理体系应该切实形成文件。 | The information security management system should be reliably documented. |

7. 关系过程 RELATIONSHIP PROCESSES

7.1. 总则 General

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| 关系过程描述了供应商管理与业务关系管理两个方面。本标准描述了服务提供方在向服务提供商提供产品 | Relationship processes describe the two related aspects of Supplier Management and Business Relationship Management. This standard addresses the role of a service |
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或服务的供应商和接受服务的客户之间的职责。

对于服务提供方而言，供应商和客户可能是内部或外部的。外部关系管理可采用正式合同形式约定，内部关系管理则通过服务或内部支持协议（通常称为运行等级协议）约定。

图2是关系简述图。

provider, who logically fills a role between suppliers, delivering goods or services to the service provider, and the customer who receives services.

Both the suppliers and customer may be internal or external to the service provider's organization. External relationships will be formalized via a contract. Internal relationships will be formalized by a service or internal under-pinning agreement often referred to as an operational level agreement.

Figure 2 shows a simplified representation of the relationships.

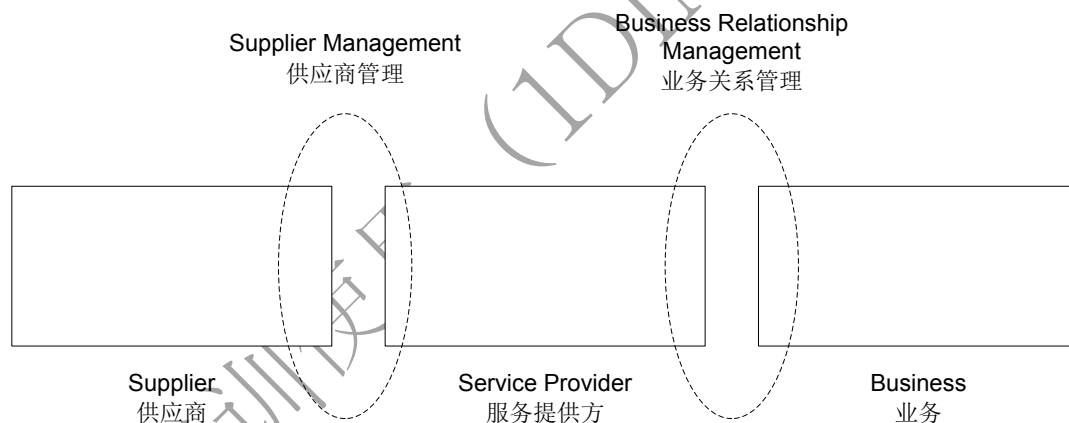


图2 关系过程 Figure 2 - Relationship processes

图2描述了服务提供商在供应链中承担的角色，供应链中的每一个步骤都是增值的过程，不论是服务提供方从供应商接受服务或商品，还是服务提供方交付增值服务给客户。

为了清晰起见，本条款中的术语“服

As Figure 2 shows, the service provider fills a role within a supply chain, where each step in the chain should be adding benefit, with the service provider receiving services or goods from the supplier and delivering an enhanced service to the customer.

For clarification, within this section the term service provider

务提供方”均指本标准所描述的
组织，无论其在供应链中承担什么角
色，也无论其参与哪些过程。

事实上，关系并非如此简单，而是包
含多个角色，同一个角色可能是供应
商，也是客户，他们之间直接或间接
通过服务提供商保持业务联系。

关系流程应确保所有相关方：

- a) 理解并满足业务需求；
- b) 了解能力和约束条件；
- c) 了解职责和责任。

同样应该确保客户满意水平是合适
的，并且未来的业务要求得到了沟通
和理解。

应该定义业务关系和供应商关系的
范围、角色和职责，并达成一致。还
应该包括利益相关方的识别、合同以
及交流的方式、频次。

is always used to describe the organization addressed by
this document, irrespective of the role, or direction in the
chain, that take in the process being described.

In practice relationships will rarely be this simple, but
comprise multiple players, taking roles both as suppliers
and customers and with business connections between
many of them directly, as well as via the service provider.

The relationship processes should ensure that all parties

- a) understand and meet business needs;
- b) understand capabilities and constraints;
- c) understand responsibilities and obligations.

They should also ensure that customer satisfaction levels
are appropriate and that future business needs are
communicated and understood.

The scope, roles and responsibilities of the business
relationship and the supplier relationship should be defined
and agreed. This should include the identification of the
stakeholders, contacts and the lines and frequency of
communication.

7.2. 业务关系管理 Business relationship management

目标：在理解客户及其业务驱动的基
础上，在服务提供方与客户之间建立
和维护良好的关系。

Objective: To establish and maintain a good relationship
between the service provider and customer based on
understanding the customer and their business drivers.

7.2.1. 服务评审 Service reviews

服务提供商和客户应该至少每年进行一次服务评审，在重大服务变更的前后也应进行评审。评审应该考虑过去的绩效、讨论当前以及计划中的业务需求，并提出服务范围和服务级别协议的任何变更。可能还应该邀请其他相关利益人，如分包方、客户、用户或其他机构代表参加评审。

The service provider and customer(s) should hold service reviews, at least annually and before and after major changes. The review should consider past performance, discuss current and projected business needs and propose any changes to the service scope and SLAs. Other stakeholders, e.g. sub-contractors, customers, user groups or other representative bodies may be invited to attend review meetings.

服务提供商和客户应该商定讨论进展、成果或问题的临时性评审过程。应该制定评审会议日程计划，并通知相关方。

The service provider and customer(s) should also agree on interim review procedures to discuss progress, achievements and issues. These meetings should be scheduled and notified to relevant stakeholders.

服务提供商应该计划并记录所有正式会议、分发会议记录并跟踪商定的措施。

The service provider should plan and record all formal meetings, issue records and follow up agreed actions.

服务提供商应该与客户建立互动关系，来了解客户的需求和重大变更，以便依据要求准备响应。

The service provider should establish a relationship with their customer such that they would expect to be aware of business needs and major changes and able to prepare to respond to that need.

7.2.2. 服务投诉 Service complaints

服务提供方和客户应该商定正式投诉程序，清晰定义什么是投诉和怎样处理投诉。服务提供方应该根据相应程序，采取合适措施以处理问题。

The service provider and customer(s) should agree on a formal complaints procedure so that there is no ambiguity on what constitutes a complaint and how it should be handled. The service provider should operate a process for

	taking appropriate action to address issues.
投诉过程应该定义服务提供方用于接受正式投诉联络方式。	The process should identify the service provider contact for formal complaints.
服务提供商应该记录、调查、采取措施、报告并关闭所有服务投诉。	The service provider should record, investigate, act upon, report and formally close all service complaints.
应该定期评审未解决的投诉，并当投诉无法在与客户约定的时间内解决时，应该提交给上级管理人员。	Outstanding complaints should be reviewed regularly and escalated to higher management if not resolved within time deadlines agreed with the customer(s).
服务提供商应该定期分析投诉记录，以识别发展趋势，并将分析结果向客户报告。	Service providers should periodically analyse the record of complaints to identify trends and report this analysis to customers.
适当时，分析结果将作为服务改进计划的输入。	The results of such analysis should be used where appropriate to inform a plan for improving the service.
7.2.3. 客户满意度测量 Customer satisfaction measurement	
应该测量客户满意度，以便服务提供方将结果与客户满意度目标及以前调查结果进行比较。应该设计合适的调查范围和复杂度，以便客户能很容易地给出反馈意见，而不用花费太多时间去精确完成调查问卷。	Customer satisfaction should be measured to enable the service provider to compare performance with customer satisfaction targets and previous surveys. The scope and complexity of the survey should be designed so customers can respond easily and without excessive time being required to complete the survey accurately.
应该调查客户满意度中的一些显著变化，并了解其原因。只有满意度调查问题和抽样方法可比时，才进行	Significant variations in satisfaction levels should be investigated and the reasons understood. Trends or other comparisons should only be made on comparable

趋势分析或其他比较。	satisfaction questions and across comparable sampling methods.
应该就客户满意度调查的结果和结论与客户进行讨论。并商定纠正措施计划，输入到服务改进计划中，并向客户报告进展情况。	The results and conclusions of customer satisfaction surveys should be discussed with the customer. An action plan should be agreed, input to a plan for improving the service and progress reported back to the customer.
对于服务的赞扬应形成文件，并向服务交付小组报告。	Compliments about the service should be documented and reported to the service delivery team.

7.3. 供应商管理 Supplier management

目标：管理供应商，以确保提供无缝的、高质量的服务。	Objective: To manage suppliers to ensure the provision of seamless, quality services.
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7.3.1. 介绍 Introduction

供应商管理程序应该确保：	Supplier management procedures should ensure that:
a) 供应商了解其对服务提供商承担的责任；	a) the supplier understands their obligations to the service provider;
b) 法定义务和商定的要求，满足商定的服务等级和范围；	b) legitimate and agreed requirements are met within agreed service levels and scope;
c) 变更得以管理；	c) changes are managed;
d) 记录与相关各方的业务交易；	d) business transactions between all parties are recorded;
e) 观察所有供应商的业绩并采取相应措施。	e) information on performance of all suppliers can be observed and acted upon.

7.3.2. 合同管理 Contract management

服务提供方应该指派一位管理者负责与供应商的合同和协议。当有多人负责此事时，应该建立统一流程，以了解供应商业绩并采取相应措施。在服务提供方内部应该为每一服务提供商指定联系人，负责与各供应商关系。

The service provider should appoint a manager responsible for contracts and agreements with suppliers. Where a number of staff are engaged in this task, there should be a common process to ensure that information on supplier performance is observed and acted on. There should be a defined contact within the service provider responsible for the relationship with each supplier.

所有供应商合同都应该包括评审时间表，以评估服务采购的业务目标是否继续有效。

All supplier contracts should contain a review schedule to assess whether the business objectives for sourcing a service remain valid.

应建立清晰定义的管理每一合同的过程。合同的修订过程也必须清晰定义。对合同管理程序的所有变更都应该正式通知所有受影响的供应商。

There should be a clearly defined process for managing each contract. The process for contract amendment should also be clearly defined. Any changes to this procedure should be formally notified to all affected suppliers.

应该在各自的组织内（服务提供商和供应商）维护联络清单。如果合同包含奖惩条款，其执行依据应该清晰表述并符合报告的要求。

A list of contact points within the respective organizations (suppliers' and service provider's) should be maintained. If a contract includes penalties or bonuses, their basis should be clearly stated and compliance to the requirements reported upon.

7.3.3. 服务定义 Service definition

服务提供商应该为每项服务和供应商维护：

For each service and supplier the service provider should maintain:

a) 服务、角色、责任的定义；

a) a definition of services, roles and responsibilities;

- | | |
|------------------------|--|
| b) 服务范围; | b) service scope; |
| c) 合同管理流程、授权级别和合同结束计划; | c) contract management process, the authorization levels and a contract exit plan; |
| d) 相关支付条款; | d) payment terms if relevant; |
| e) 商定报告的内容和服务绩效记录。 | e) agreed reporting parameters and records of achieved performance. |

7.3.4. 管理多供应商 Managing multiple suppliers

不管是服务提供方直接与所有供应商交易, 还是由主供应商负责与各分包供应商联系, 管理责任应该清晰。

It should be clear whether the service provider is dealing with all suppliers directly or a lead supplier taking responsibility for sub-contracted suppliers.

主供应商应该记录所有分包供应商的名字、责任和关系, 并按需提供给服务提供方。

The lead supplier should record the names, responsibilities and relationships between all sub-contracted suppliers, and make this available to the service provider if required.

服务提供方应该获得主供应商正式管理分包供应商的证据, 如果合适, 应根据ISO/IEC 20000-1管理分包供应商。

The service provider should obtain evidence that lead suppliers are formally managing sub-contracted providers, guided, where appropriate, by the requirements in ISO/IEC 20000-1.

7.3.5. 合同争议管理 Contractual disputes management

服务提供方和供应商都应该建立管理合同争议的过程, 并且该过程应该在合同中定义或引用。

Both the service provider and the supplier should operate a process for managing disputes and this should be defined or referred to within the contract.

如果争议无法通过正常途径解决时,

An escalation route should be available for disputes that

应该有升级的途径。

cannot be resolved through the normal route.

该过程应该确保争议被记录、调查、解决并正式关闭。

The process should ensure that disputes are recorded, investigated, acted upon and formally closed.

7.3.6. 合同终止 Contract end

合同管理流程应该包括合同结束条款，无论是正常结束还是提前结束。还应该包括将服务转移给第三方的内容。

The contract management process should include provision for the end of contract – either expected end or early end. It should also provide for the transfer of the service to another organization.

8. 解决过程 RESOLUTION PROCESSES

8.1. 背景 Background

事故和问题管理是独立过程，虽然它们紧密相关。事件管理关注的是为用户服务恢复，而问题管理考虑的是识别并消除事件的根本原因。

Incident and problem management are separate processes although they are closely linked. Incident deals with the restoration of service to the users, whereas problem is concerned with identifying and removing the causes of incidents.

8.1.1. 设置优先级 Setting priorities

应该根据优先级确定解决的目标。优先级的确定应该根据影响和紧急程度。影响应该根据对客户业务的实际或潜在损害程度。紧急程度应该根据问题或事件被发现到客户业务受损的时间间隔。

Targets for resolution should be based on priority. Priority should be based on impact and urgency. Impact should be based on the scale of actual or potential damage to the customer's business. Urgency should be based on the time between the problem or incident being detected and the time that the customer's business is impacted.

事件或问题解决的时间安排至少应考虑以下因素：

- a) 优先级；
- b) 可用的技能；
- c) 资源要求；
- d) 解决方案的效益/成本；
- e) 实施解决方案所需时间。

注：优先级贯穿于这个服务管理，是事件管理和问题管理的核心。

The scheduling of incident or problem resolution should take into account at least the following:

- a) priority;
- b) skills available;
- c) competing requirements for resources;
- d) effort/cost to provide the method of resolution;
- e) elapsed time to provide a method of resolution.

NOTE Priority is used throughout service management but is central to incident and problem management.

8.1.2. 替代方案 Workarounds

适当时，问题管理应该建立并维护替代方案，使得事件管理可以帮助用户工作人员恢复服务。

只有当成功实施纠正性的变更，或错误不再存在（如不再使用该服务）时，已知错误才应该被关闭。

问题管理应该有权访问问题所涉及的业务领域信息。

存放在知识库中的替代方案的信息，其适用性和有效性应该予以维护。

Where appropriate, problem management should develop and maintain workarounds to enable incident management to help service restoration by users of staff.

A known error should only be closed where a corrective change has been successfully applied, or the error is no longer applicable, e.g. because the service is no longer used.

Problem management should have access to information on the business areas affected by problems.

Information on workarounds stored in the knowledge base, their applicability and effectiveness should be stored and

maintained.

8.2. 事件管理 Incident management

目标：尽可能快的恢复商定的业务服务或响应服务请求。

Objective: To restore agreed service to the business as soon as possible or to respond to service requests

8.2.1. 总则 General

注1：事件管理过程可能由负责日常与用户联系的服务台提供。

NOTE 1 The incident management process may be delivered by a service desk, which acts as the day-to-day contact with the users.

注2：事件管理应该是：

NOTE 2 Incident management should be:

a) 影响服务或可能影响服务的事件的预防或被动响应过程；

a) both a proactive and reactive process, responding to incidents that affect, or potentially could affect the service;

b) 关注客户服务的恢复，而不是判断事件原因。

b) concerned with the restoration of the customers' service, not with determining the cause of incidents.

事件管理流程应该包括：

The incident management process should include the following:

a) 接受保修、并记录、分配优先级和分类；

a) call reception, recording, priority assignment, classification;

b) 事件的一线决或转发；

b) first line resolution or referral;

c) 安全问题的考虑；

c) consideration of security issues;

d) 事件跟踪和生命周期管理；

d) incident tracking and lifecycle management;

e) 事件的验证和关闭;

e) incident verification and closure;

f) 客户的一线联系;

f) first line customer liaison;

g) 事件的升级。

g) escalation.

可以通过电话、语音邮件、来访、书信、传真或者电子邮件报告事件, 用户也可以直接访问事件记录系统进行事件记录, 或者采用自动监控软件报告事件。

Incidents may be reported by telephone calls, voice mails, visits, letters, faxes or e-mails, or may be recorded directly by users with access to the incident recording system, or by automatic monitoring software.

应该以可检索和分析的方式, 记录所有事件。

All incidents should be recorded in a manner that allows relevant information to be retrieved and analysed.

应该与实际受影响或潜在影响的相关人员沟通事件解决进展(或没有进展)。

Progress (or lack of it) in resolving incidents should be communicated to those actually or potentially affected.

所有相关活动都应该记录在事件记录中。

All actions should be recorded on the incident record.

事件管理人员应该有权访问最新的知识库。知识库中包括技术专家、以前的事件、相关问题和已知错误、替代方案、检查表等有助于恢复服务的各种信息。

Incident management staff should have access to an up-to-date knowledge base holding information on technical specialists, previous incidents, related problems and known errors, workarounds and checklists that will help in restoring the service to the business.

只要可能, 应该向客户提供让业务得以持续的方法, 即使只是一个被降级的服务, 例如禁止已出错的功能。其

Wherever possible, the customer should be provided with the means to continue business, even if only with a degraded service, e.g. by disabling a faulty feature. The

目的就是最大限度降低对客户业务活动的影响。当未找出的原因，但已有替代方法的，应该记录详细信息以用于进行中的问题分析或类似事件再次发生。

只有当事件报告用户证实事件已经解决并且服务已经恢复的时候，事件才能正式关闭。

motive is to minimize the impact on the customer's business activities. When the cause remains undiagnosed but a workaround is established, details should be recorded for use during continuing problem diagnosis, and when similar incidents recur.

Final closure of an incident should only take place when the initiating user has been given the opportunity to confirm that the incident is now resolved and service restored.

8.2.2. 重大事件 Major incidents

应该清晰定义什么构成一个重大事件，以及谁被授权可以对事件或问题处理过程进行变更。

在任何时候所有重大事件都应有一个明确的管理者负责。

重大事件经理应该向协调和控制事件的所有方面的角色适当授权。其中包括事件升级，以及与参与解决的相关各方和受重大事件影响的客户进行沟通。

注：授权可以是暂时的，而且只有在出现重大事件时有效。

重大事件处理流程应包括评审，评审

There should be a clear definition of what constitutes a major incident and who is empowered to invoke changes to the normal operation of the incident/problem process.

All major incidents should have a clearly defined responsible manager at all times.

Nomination as manager of a major incident should give the individual authority levels that are adequate to the role of coordinating and controlling all aspects for the resolution. This should include the responsibility for effective escalation and communication across all areas involved in resolution, and to the customers affected by the major incident.

NOTE This level of authority can be temporary, and apply only during that major incident.

The process for a major incident should include a review

将作为服务改进计划的输入。

which will inform a plan for improving the service.

8.3. 问题管理 Problem management

目标：通过对事件起因的预先识别和分析，以及管理问题的关闭，以最大限度降低对业务的损害。

Objective: To minimize disruption to the business by proactive identification and analysis of the causes of incidents and by managing problems to closure.

8.3.1. 问题管理的范围 Scope of problem management

问题管理过程应该调查事件的根本原因。

The problem management process should investigate the underlying causes of incidents.

问题管理应该根据业务要求预防事件或已知错误重复发生或再现。

Problem management should proactively prevent the recurrence or replication of incidents or known errors according to the business requirements.

8.3.2. 启动问题管理 Initiation of problem management

应该对事件进行分类，以帮助确定问题原因。分类参考现有问题和变更。

Incidents should be classified to help determine the causes of problems. Classification may reference existing problems and changes.

注：早期登录的事件分类可能还会受到其它因素的影响，例如服务、受影响业务领域和症状等。

NOTE On initial registration incidents categorisation will be influenced by other factors also including service, business area affected and symptoms presented.

8.3.3. 已知错误 Known errors

当问题管理调查已经明确事件的根本原因和解决事件方法时，该问题应该被归类为已知错误。

When the problem management investigation has identified the root cause of an incident and a method of resolving the incident, the problem should be classified as a known error.

除了在故障时被怀疑的配置项，所有已知错误还有应该对照受影响的服务或潜在影响的服务加以记录。

All known errors should be recorded against the current and potentially affected services in addition to the configuration item suspected of being at fault.

在已导入运行环境的服务中的已知错误的信息应该传递给服务管理方，并连同替代方案记录在知识库中。已知错误在成功解决后才应该被关闭。

Information on known errors in services being introduced into the live environment should be passed to service management and should be recorded in the knowledge base, together with any workarounds. A known error should not be closed until after successful resolution.

注： 客户或服务提供方可能认为解决方案开支太大，或者对业务无益。此时，应该清晰的形成文件。已知错误记录保持未关闭状态，因为事件的结果可能依然存在，并可能需要替代方法或需要重新评估问题解决的决策。

NOTE The customer or service provider may decide that the resolution is too expensive or not of benefit to the business. If this is the case it should be clearly documented. The known error record should remain open however, since consequential incidents are likely to still occur and may require workarounds and/or require reassessment of the decision to resolve.

8.3.4. 问题的解决 Problem resolution

当根本原因得以识别，并对解决方案做出了决策，解决方案应通过变更管理过程来实施。

When the root cause has been identified, and a decision to resolve it has been made, the resolution should be progressed via the change management process.

8.3.5. 沟通 Communication

替代方法、永久性方案或问题的进展，应该与受影响或要求提供支持的相关各方沟通。

Information on workarounds, permanent fixes or progress of problems should be communicated to those affected or required to support affected services.

8.3.6. 跟踪和升级 Tracking and escalation

应该跟踪所有问题的进展。

The progress of all problems should be tracked.

所有议题应该升级给合适的相关方。

All issues should be escalated to the appropriate parties.

跟踪和升级过程应该包括：

The process should cover:

a) 记录问题生命周期内问题解决
责任人的变更；

a) recording changes to the identities of those responsible
for problem resolution during the lifecycle of each
problem;

b) 识别违背服务等级目标的事件；

b) identification of incidents that breach service level
targets;

c) 让客户和同事了解相关信息，以
便他们采取相关合适行动将问
题影响降到最低；

c) cascading information to customers and colleagues so
that they can take appropriate action to minimize the
impact of the unresolved problem;

d) 定义问题的升级点；

d) defining the escalation points;

e) 记录采用的资源和所采取的行
动。

e) the recording of the resources used and any actions
taken.

8.3.7. 事件和问题记录关闭 Incident and problem record closure

记录关闭程序应该包括检查以确保：

The record closure procedure should include checking to
ensure that:

a) 解决的信息被准确记录；

a) details of resolution have been accurately logged;

b) 对原因进行了分类以便分析；

b) the cause is categorized to facilitate analysis;

c) 适当时，让客户和员工了解解决
方案；

c) if appropriate, both the customer and support staff are
aware of the resolution;

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| d) 客户同意的解决方案已经被实施； | d) the customer agrees that the resolution has been achieved; |
| e) 如果解决方案没有成功或者无法实现，应告知客户； | e) if a resolution is not to be achieved or not possible, the customer is informed. |

8.3.8. 问题评审 Problem reviews

在调查尚未解决的问题、非常见问题或者影响重大的问题的时，应该进行问题评审。其目的是寻找过程的改进计划，防止事件或错误的再次发生。

Problem reviews should be held where investigation into unresolved, unusual or high-impact problems justifies them. Their purpose is to seek improvements to the process and to prevent recurrence of incidents or mistakes.

问题评审通常：

Problem reviews are typically:

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| a) 依据服务等级评审每个事件的等级和问题的状态； | a) reviews of individual incident levels and problem status against service levels; |
| b) 管理评审以强调需要立即采取行动的问题； | b) management reviews to highlight those problems that require immediate action; |
| c) 管理评审以确定并进行趋势分析，以作为其他过程的输入，如用户教育和培训。 | c) management review to determine and analyse trends and to provide input for other processes, such as user education and training. |

8.3.9. 评审事项 Topics for reviews

评审应该包括识别：

The reviews should include identification of:

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| a) 趋势，例如重复发生的问题和事件、已知错误等； | a) trends, e.g. recurring problems and incidents, known errors etc; |
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b) 某一特定类型的组件和位置，重复发生问题；	b) recurring problems of a particular classification component or location;
c) 由于资源、培训和文档导致的缺陷；	c) deficiencies caused by resourcing, training or documentation;
d) 不符合；如违背标准、方针和法规；	d) non-conformances, e.g. against standards, policies and legislation;
e) 计划版本中的已知错误；	e) known errors in planned releases;
f) 在解决事件或问题中承诺的人员资源；	f) staff resource commitment in resolving incidents and problems;
g) 已解决问题或事件的重复发生；	g) recurrence of resolved incidents or problems.
应该记录对服务或问题管理过程的改进，并作为服务改进计划的输入。	Improvements to the service or the problem management process should be recorded and fed into a plan for improving the service.
相关信息应加入问题管理知识库。	The information should be added to the problem management knowledge base.
应该对相关文件进行更新，如用户指南和系统文件。	All relevant documentation should be updated, e.g. user guides and system documentation.

8.3.10. 问题预防 Problem prevention

预防性的主动问题管理可以促进事件和问题的减少。其中包括提供辅助分析的信息，例如：	Proactive problem management should lead to a reduction in incidents and problems. It should include reference to information that assists analysis, such as:
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| a) 资产和配置; | a) asset and configuration; |
| b) 变更管理; | b) change management; |
| c) 已发布的已知错误, 从供应商获得的替代方法; | c) published known error, workaround information from suppliers; |
| d) 相似问题的历史信息。 | d) historical information on similar problems. |

问题预防包括对个别事件的预防(例如某系统的一个特性反复出现的问题)到战略决策。后者要求较大投入, 例如筹建更好的网络, 此时问题管理可合并为可用性管理。

Problem prevention should range from prevention of individual incidents, such as repeated difficulties with a particular feature of a system, through to strategic decisions. The latter can require major expenditure to implement such as investment in a better network, at this level proactive problem management merges into availability management.

问题预防还包括向客户提供信息, 以便他们将来不再寻求帮助, 例如预防一些由于用户缺乏知识和培训导致的事件。

Problem prevention also includes information being given to customers that means they do not need to ask for assistance in the future, e.g. preventing incidents caused by lack of user knowledge or training.

9. 控制过程 Control processes

9.1. 配置管理 Configuration management

目标: 定义并控制服务和基础设施的组件, 并维护准确的配置信息。

Objective: To define and control the components of the service and infrastructure and maintain accurate configuration information.

9.1.1. 配置管理计划和实施 Configuration management planning and implementation

配置管理应结合变更和发布管理进行计划和实施，以确保服务提供方可以有效管理其IT资产和配置。

Configuration management should be planned and implemented with change and release management to ensure that the service provider can manage its IT assets and configurations effectively.

准确的配置信息应该为新建或变更的服务和系统的版本发布和分发计划及变更控制提供支持。目的是形成一个整合服务提供方的配置信息过程及包括客户和供应商（必要时）的有效体系。

Accurate configuration information should be available to support the planning and control of changes as new and updated services and systems are released and distributed. The result should be an efficient system that integrates the service provider's configuration information processes and those of its customers and suppliers, where appropriate.

应该考虑所有重要资产和配置，并指派管理责任人，以确保维护合适的保护和控制，例如变更在实施前实施授权。

All major assets and configurations should be accounted for and have a responsible manager who ensures that appropriate protection and control is maintained, e.g. changes are authorized before implementation.

控制措施实施的职责可以委托给其他人员，但管理责任人仍然负有责任。应该向管理责任人提供履行职责的必要信息，如，进行变更授权时，可能需要变更的成本、风险和影响，以及和实施所需资源等信息。

Responsibility for implementing controls may be delegated but accountability should remain with the responsible manager. The responsible manager should be provided with the information necessary to discharge this responsibility, e.g. the person authorising a change may require information on the cost, risks, impact of the change and resources for implementation.

基础设施和（或）服务应该有更新的配置管理计划，该计划可能是一个单独文件，也可能是作为其它计划文件的一部分。该计划应该包括或描述：

The infrastructure and/or services should have up-to-date configuration management plan(s) that may be stand-alone or form part of other planning documents. They should include or describe:

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| a) 范围、目标、方针、标准的角色和责任； | a) scope, objectives, policies, standards roles and responsibilities; |
| b) 定义服务和基础设施的配置项、控制配置变更、记录和报告配置项状态以及验证配置项的完整性和准确性的配置管理过程； | b) the configuration management processes to define the configuration items in the service(s) and infrastructure, control changes to the configurations, recording and reporting the status of configuration items and verifying the completeness and correctness of configuration items; |
| c) 对于可问责、可追溯、可审计的要求，如来自安全、法律、法规或业务目的的要求； | c) the requirements for accountability, traceability, auditability, e.g. for security, legal, regulatory or business purposes; |
| d) 配置控制（访问、保护、版本、开发、版本控制）； | d) configuration control (access, protection, version, build, release controls); |
| e) 识别、记录和管理两个或多个组织的公共边界的配置项及信息的接口控制过程，如系统接口和版本； | e) interface control process for identifying, recording, and managing the configuration items and information at the common boundary of two or more organizations, e.g. system interfaces, releases; |
| f) 资源管理计划和建立，以便将资产和配置置于控制之下，并维持配置管理系统，如培训； | f) planning and establishing the resources to bring assets and configurations under control and maintain the configuration management system, e.g. training; |
| g) 管理供应商和分包商的配置管理； | g) management of suppliers and sub-contractors performing configuration management. |

注：可采取一定自动化手段，以确保流程不会无效、错误百出、甚至失效。	NOTE An appropriate level of automation should be implemented to ensure that processes do not become
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either inefficient, error prone or may not be followed at all.

9.1.2. 配置识别 Configuration identification

所有配置项应该被唯一标识，并定义描述其功能和物理特性的属性。信息应是相关，并可审计的。

All configuration items should be uniquely identified and defined by attributes that describe their functional and physical characteristics. Information should be relevant and auditable.

应该采用合适的标记或其它识别方法，并将其记录在配置管理数据库中。

Appropriate markings, or other methods of identification, should be used and recorded in the configuration management database.

应该利用已建立的选择标准来管理配置项，并应该包括：

Items to be managed should be identified using established selection criteria and should include:

a) 信息系统、软件（包括第三方软件）和相关系统文件的所有议题和版本，如，要求、规范、设计、测试报告、版本文件；

a) all issues and releases of information systems and software (including third-party software) and related system documentation, e.g. requirements specifications, designs, test reports, release documentation;

b) 为适用的环境、标准硬件架构和版本建立配置基准或架构的描述；

b) configuration baselines or build statements for each applicable environment, standard hardware builds and release;

c) 主拷贝和电子资料库，如最终软件库；

c) master hardcopy and electronic libraries, e.g. definitive software library;

d) 使用的配置管理包或工具；

d) configuration management package or tools used;

e) 许可证；

e) licences;

f) 安全组件，如防火墙；	f) security components, e.g. firewalls;
g) 出于财务管理和业务原因需要跟踪的物理资产，如安全磁盘介质、设备；	g) physical assets that need to be tracked for financial asset management or business reasons, e.g. secure magnetic media, equipment;
h) 服务相关文件，如SLA、程序；	h) service related documentation, e.g. SLAs, procedures;
i) 服务支持设施，例如机房电源；	i) service supporting facilities, e.g. power to computer room;
j) 配置项之间的关系和依赖性；	j) relationships and dependencies between configuration items.
注：其它可视为配置项的包括：	NOTE Other items that may be considered as configuration items include:
a) 其它文档；	a) other documentation;
b) 其它资产；	b) other assets;
c) 其它设施，例如站点；	c) other facilities, e.g. site;
d) 业务单元；	d) business units;
e) 人员。	e) people.
应该识别配置项之间适当的关系和依赖性，以便提供必要的控制等级。	Appropriate relationships and dependencies between configuration items should be identified to provide the necessary level of control.
如果需要跟踪配置项，那么应该确保	Where traceability is required the process should ensure

从需求文档到版本发布记录整个生命周期内都是可追踪的，如实施可追溯矩阵。	that configuration items can be traced through the full lifecycle, from requirements documents through to release records, e.g. using a traceability matrix.
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9.1.3. 配置控制 Configuration control

配置过程应该确保只有经过授权且可识别的配置项才被接受，并记录从接受到销毁的全过程。	The process should ensure that only authorized and identifiable configuration items are accepted and recorded from receipt to disposal.
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没有适当的控制文件（如经过批准的变更请求、更新的版本信息），不能添加、修改、替换/删除配置项。	No configuration item should be added, modified, replaced or removed/withdrawn without appropriate controlling documentation, e.g. approved change request, updated release information.
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为保护系统、服务和基础设施的完整性，配置项应该保存在一个合适的安全环境：	To protect the integrity of systems, services and the infrastructure, configuration items should be held in a suitable and secure environment which:
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| a) 保护其不受未经授权的访问、变更或破坏，例如病毒； | a) protects them from unauthorized access, change or corruption, e.g. virus; |
| b) 提供灾难后的恢复方法； | b) provides a means for disaster recovery; |
| c) 允许对受控备份的受控恢复，如软件。 | c) permits the controlled retrieval of a copy of the controlled master, e.g. software. |

9.1.4. 配置状态说明和报告 Configuration status accounting and reporting

应该维护当前准确的配置记录，以反映配置项状态、位置和版本的变更。	Current and accurate configuration records should be maintained to reflect changes in the status, location and versions of configuration items.
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状态清单应该提供配置项在整个生命周期中的当前和历史数据。应该利用多种状态描述使得配置项可以被跟踪，如订购、接收、验收测试、上线、变更中、撤回、销毁等。

配置项信息应该保持更新，并可以被已定义配置项变更的计划、决策和管理所用。

如果需要，用户、客户、供应商和合作方应该可以访问配置信息，以协助其制定计划和决策。例如，外部服务提供商可以提供配置信息给客户和其他相关各方，以支持其他服务管理过程的端到端服务。

配置项管理报告应该提供给所有相关各方。报告应该包括配置项的识别和状态、版本和相关文件。报告应该包括：

- a) 配置项最新版本；
- b) 配置项的位置和软件主版本的位置；

Status accounting should provide information on the current and historical data concerned with each configuration item throughout its lifecycle. It should enable changes to configuration items to be tracked through various states, e.g. ordered, received, in acceptance test, live, under change, withdrawn, disposed.

Configuration information should be kept current and made available for planning, decision making and managing changes to the defined configurations.

Where required, configuration information should be accessible for users, customers, suppliers and partners to assist them in their planning and decision making. For example, an external service provider may make configuration information accessible to the customer and other parties to support the other service management processes for the end-to-end service.

Configuration management reports should be available to all relevant parties. The reports should cover the identification and status of the configuration items, their versions and associated documentation. Reports should cover:

- a) latest configuration item versions;
- b) location of the configuration item and for software the location of the master versions;

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| c) 相互依赖关系; | c) interdependencies; |
| d) 版本历史; | d) version history; |
| e) 配置项状态及构成: | e) status of configuration items that together constitute: |
| 1) 服务配置或系统; | 1) service configuration or system; |
| 2) 变更、基准、编译或版本; | 2) a change, baseline, build or release; |
| 3) 版本号或变量; | 3) version or variant. |

9.1.5. 配置验证和审计 Configuration verification and audit

应该计划配置验证和审计过程（包括物理的和功能性的），并检查是否有充分的流程和资源：

Configuration verification and audit processes, both physical and functional, should be scheduled and a check performed to ensure that adequate processes and resources are in place to:

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| a) 保护物理配置和组织知识资产; | a) protect the physical configurations and the intellectual capital of the organization; |
| b) 确保服务提供方控制其配置、主拷贝和许可证; | b) ensure that the service provider is in control of its configurations, master copies and licences; |
| c) 确保配置信息是准确、可控和可见的; | c) provide confidence that configuration information is accurate, controlled and visible; |
| d) 确保变更、版本、系统或环境符合其合同约定或指定的要求，而且配置记录是准确的。 | d) ensure that a change, a release, a system or an environment conforms to its contracted or specified requirements and that the configuration records are accurate. |

应该定期进行配置审计，此外还应该在重大变更前后、灾难后和随机的不定期进行审计。

Configuration audits should be carried out regularly, before and after major change, after a disaster and at random intervals.

缺陷或不符之处应该被记录、评估和改进，反馈意见应提供给相关各位和服务改进项目；

Deficiencies and non-conformities should be recorded, assessed and corrective action initiated, acted upon and fed back to the relevant parties and plan for improving the service.

注：通常存在两种类型的配置审计：

NOTE Normally there are two types of configuration audits:

- a) 功能性配置审计：了解配置项是否达到配置文档事先确定的性能和功能特性的正式检查；
- a) functional configuration audit: a formal examination to verify that a configuration item has achieved the performance and functional characteristics specified in its configuration documents;
- b) 物理配置审计：了解配置项的“当前编译版/产品版”是否与其产品配置文件相符的正式检查。
- b) physical configuration audit: a formal examination of the “as-built/produced” configuration of a configuration item to verify that it conforms to its product configuration documents.

9.2. 变更管理 Change management

目标：确保以受控的方式去评估、批准、实施和评审所有的变更。

Objective: To ensure all changes are assessed, approved, implemented and reviewed in a controlled manner.

9.2.1. 计划和实施 Planning and implementation

变更管理过程和程序应该确保：

The change management processes and procedures should ensure that:

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| a) 在文件中清晰定义变更的范围； | a) changes have a clearly defined and documented scope; |
| b) 只有为业务带来利益的变更才能被认可，例如商业的、法律的、规章的、法令的； | b) only changes that provide business benefit are approved, e.g. commercial, legal, regulatory, statutory; |
| c) 根据优先级和风险为变更安排时间； | c) changes are scheduled based on priority and risk; |
| d) 配置的变更可以实施过程中验证； | d) changes to configurations can be verified during change implementation; |
| e) 对变更的时间进行监视，如果需要应该进行改进； | e) the time to implement changes is monitored and improved where required |
| f) 可以证实，变更是如何： | f) it can be demonstrated how a change is: |
| 1) 发起、记录和分类（参考变更发起如何变更的文件）； | 1) raised, recorded and classified (with references to documents that gave rise to the change); |
| 2) 评估变更对服务、客户和版本发布计划的影响、紧急程度、成本、收益和风险； | 2) assessed for the impact, urgency, cost, benefits and risk of the changes on the service, customer and release plans; |
| 3) 未成功时的恢复或补救； | 3) reversed or remedied if unsuccessful; |
| 4) 归档，如与受影响的配置项相关的变更请求、实施后更新的版本号和版本发布计划； | 4) documented, e.g. the change request is linked to affected configuration items and the updated version of the implementation and release plans; |

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| <p>5) 变更授权人根据变更的类型、规模和风险对变更请求的接受或拒绝；</p> <p>6) 由所变更组件的责任团队内的指定责任人负责实施；</p> <p>7) 测试、验证和签署；</p> <p>8) 关闭和评审；</p> <p>9) 时间安排、监视和报告；</p> <p>10) 适当时，与事件、问题、其它变更和配置项记录联系。</p> | <p>5) approved or rejected by a change authority, depending on the type, size and risk of change;</p> <p>6) be implemented by the nominated owner within the groups responsible for the components being changed;</p> <p>7) tested, verified and signed off;</p> <p>8) closed and reviewed;</p> <p>9) scheduled, monitored and reported on;</p> <p>10) linked to incident, problem, other change and configuration item records where appropriate.</p> |
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变更的状态和计划的实施日期应该作为变更和版本发布日程安排的依据。

The status of changes and scheduled implementation dates should be used as the basis for change and release scheduling.

日程安排信息应该提供给受变影响的更相关人员。

Scheduling information should be made available to the people affected by the change.

如果可能导致正常服务时间内的服务受损，应该在变更实施前获得受影响人员的同意。

Where an outage can be caused during normal service hours the people affected should agree to the change before implementation.

9.2.2. 关闭和评审变更请求 Closing and reviewing the change request

所有变更在变更实施后，都应该进行成功或失败的评审，并记录所有

All changes should be reviewed for success or failure after implementation and any improvements recorded. A

改进措施。重大变更实施后应该进行评审，以检查：

- a) 变更是否满足目标；
- b) 客户对结果是否满意；
- c) 没有预期之外的负面影响；

应该记录所有不符合，并采取相应改进措施。

在评审变更管理过程中，所识别的任何缺陷和弱点，应该作为服务改进计划的输入。

post-implementation review should be undertaken for major changes to check that:

- a) the change met its objectives;
- b) the customers are happy with the results;
- c) there have been no unexpected side effects.

Any nonconformity should be recorded and actioned.

Any weaknesses or deficiencies identified in a review of the change management process should be fed into plans for improving the service.

9.2.3. 紧急变更 Emergency changes

有时可能需要紧急变更，应该尽可能遵循变更流程，某些细节可以事后记录。当紧急变更流程跳过其它变更管理要求时，变更应该尽可能满足适用可行的要求。

实施人员应该判断紧急变更的必要性，并在变更后评审，以确保是紧急变更。

Emergency changes are sometimes required and where possible the change process should be followed but some details may be documented retrospectively. Where the emergency process bypasses other change management requirements, the change should conform to these requirements as soon as practicable.

Emergency changes should be justified by the implementer and reviewed after the change to verify that it was a true emergency.

9.2.4. 变更管理报告、分析和措施 Change management reporting, analysis and actions

应该定期分析变更记录，以检测不断

Change records should be analysed regularly to detect

提高的变更等级、频繁再现的类型、呈现的趋势和其它相关信息。应该记录变更分析的结果和结论，并采取相应行动。	increasing levels of changes, frequently recurring types, emerging trends and other relevant information. The results and conclusions drawn from change analysis should be recorded and acted upon.
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10. 发布过程 Release process

10.1. 发布管理过程 Release management process

目标：交付、分发和跟踪导入实际运作环境中的一个或多个变更。	Objective: To deliver, distribute and track one or more changes into the live environment.
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10.1.1. 总则 General

发布管理应该协调服务提供方、多个供应商和业务，以计划在分布的环境中提交版本。	Release management should coordinate the activities of the service provider, many suppliers and the business to plan and deliver a release across a distributed environment.
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良好的计划和管理是版本打包和成功分发以及管理其对业务和IT的相关影响和风险的基础。应结合业务对受影响的信息系统、基础设施、服务和文档的版本发布进行计划。	Good planning and management are essential to package and successfully distribute a release, and to manage the associated impact and risks to the business and IT. The release of affected information systems, infrastructure, services and documentation should be planned with the business.
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发布的版本中应该包括文件的所有相关更新，如业务过程、支持文档和服务等级协议。	All associated updates to documentation should be included in the release, e.g. business processes, support documents and service level agreements.
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所有对授权变更造成影响的新的或	The impact of all new or changed configuration items
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经过改动的配置项都应该被评估。

required to effect the authorized changes should be assessed.

服务提供方应该确保版本发布的技术或非技术两个方面都已经考虑。

The service provider should ensure that both technical and non technical aspects of the release are considered together.

版本发布项应该可跟踪并防止修改。
只有经过适当测试和认可的发布才应该被导入运行环境。

The release items should be traceable and secure from modification. Only suitably tested and approved releases should be accepted into the live environment.

10.1.2. 版本发布策略 Release policy

版本发布政策应该包括：

There should be a release policy that includes the:

a) 发布频次和类型；

a) frequency and type of release;

b) 版本发布管理的角色和责任；

b) roles and responsibilities for release management;

c) 版本导入到测试和生产环境的授权；

c) authority for the release into acceptance test and production environments;

d) 所有版本的唯一标识和描述；

d) unique identification and description of all releases;

e) 将变更组合到版本的方法；

e) approach to grouping changes into a release;

f) 为提高效率和可重复性，建立、安装、版本分发过程的自动化方法；

f) approach to automating the build, installation, release distribution processes to aid repeatability and efficiency;

g) 版本发布的验证和接受。

g) verification and acceptance of a release.

10.1.3. 发布和撤销计划 Release and roll-out planning

服务提供方应该在业务环境下工作，以确保将发布的配置项彼此兼容并与发布环境中配置项相兼容。

The service provider should work with the business to ensure that the configuration items that are to be released are compatible with each other and with configuration items in the target environment.

发布计划应该确保受影响信息系统、基础设施、服务和文件的变更得到认可、授权、计划安排、协调和跟踪。

Release planning should ensure that the changes to affected information systems, infrastructure, services and documentation are agreed, authorized, scheduled, coordinated and tracked.

发布和回退应该分阶段计划，因为回退的细节可能事先并不知道。

The release and roll out should be planned in stages as details of the roll out might not be known initially.

发布和回退计划一般应该包括：

The planning for a release and roll out should typically include:

a) 版本发布日期和可交付物的描述；

a) release dates and description of deliverables;

b) 所发布版本相关的变更、关闭或解决的问题和已知错误，以及在版本测试期间所识别的已知错误；

b) related changes, problems and known errors closed or resolved by this release and known errors that have been identified during testing of the release;

c) 实施跨越所有业务和地理单元版本发布的相关流程；

c) related processes to implement a release across all business and geographical units;

d) 如发布失败，用于回退或补救的方式；

d) the manner in which the release will be backed-out or remedied if unsuccessful;

e) 验证和验收过程；

e) verification and acceptance process;

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|--|--|
| f) 与客户和服务支持人员的沟通，以及相关准备、文件的提供和培训； | f) communication, preparation, documentation and training for customer and support staff; |
| g) 采购、存储、分发、联系、接受和销毁商品的后勤工作和流程； | g) logistics and processes to purchase, store, dispatch, connect, accept and dispose of goods; |
| h) 确保维护服务等级所需的支持资源； | h) support resources required to ensure service levels are maintained; |
| i) 可能对将版本导入测试和运行环境造成影响的相关变更和风险的识别； | i) the identification of dependencies, related changes and associated risks that might affect the smooth transfer of a release into the acceptance test and production environments; |
| j) 发布结束的签署； | j) release sign off; |
| k) 如果必要，当进行重大升级时，应该安排对生产环境的审计，确保发布安装时的运行环境与预期状态一致。 | k) schedule of audits of the production environment where required for major upgrades to ensure that the live environment is in the expected state when the release is installed. |

10.1.4. 软件的开发和获取 Developing or acquiring software

应该对内部团队、系统开发商、系统集成商或者其它组织的信息系统和软件的发布版本进行验收。

Information systems and software releases from in-house teams, systems builders, system integrators or other organizations should be verified on receipt.

整个过程应该文件化到配置管理计划中。

The overall process should be documented in the configuration management plan.

10.1.5. 设备、建立和配置的发布 Design, build and configure release

应该对版本和分发进行设计和实施，以：

- a) 与服务提供方的系统架构、服务管理和基础设施标准相吻合；
- b) 确保编译、安装、处理、打包和交付的完整性；
- c) 在编译和发布过程中采用软件库和相关资源库来管理和控制组件；
- d) 清晰识别风险，并根据需要采取补救行动；
- e) 使得在安装前验证目标平台满足先决条件；
- f) 使得当发布实现目标时，验证发布的完整性。

本过程的结果包括版本通告、安装指南、基于相关配置基准线的已安装软硬件。

版本发布的结果应该提交给测试小组。

可以利用自动化系统编译、安装、发

Release and distribution should be designed and implemented to:

- a) conform with the service provider's systems architecture, service management and infrastructure standards;
- b) ensure the integrity is maintained during build, installation, handling, packaging and delivery;
- c) use software libraries and related repositories to manage and control components during the build and release process;
- d) risks are clearly identified and remedial action can be taken if required;
- e) enable verification that the target platform satisfies prerequisites before installation;
- f) enable verification that a release is complete when it reaches its destination.

The outputs from this process should include release notes, installation instructions, installed software and hardware with related configuration baseline.

The outputs from the release should be handed over to the group responsible for testing.

Build, installation, release and distribution processes might

布和分发流程以减少出错，确保流程的可重复性和确保新版本能够快速发布。

be automated to reduce errors, ensure that the process is repeatable and that new releases can be rolled out quickly.

10.1.6. 发布验证和接受 Release verification and acceptance

最终的结果应该是对照要求的、完整的发布包的结束签署。

The end result should be a sign-off on completeness of the whole release package against requirements.

验证和接受过程包括：

The verification and acceptance processes should:

a) 验证受控的验收测试环境是否符合目的生产环境的要求；

a) verify that the controlled acceptance test environment matches the requirements of target production environment;

b) 确保发布版本是基于配置管理下的版本创建的，并采用事先计划好的工作流程安装到验收测试环境；

b) ensure that the release is created from versions under configuration management and installed in the acceptance test environment using the planned production process;

c) 检查测试已经完成，例如功能性测试和非功能性测试，业务许可测试，开发、发布、分发和安装规程的测试；

c) verify that the appropriate level of testing has been completed, e.g. functional and non-functional testing, business acceptance testing, testing of the build, release, distribution and installation procedures;

d) 确保发布版本经过测试，以使客户和服务提供人员满意；

d) ensure that the release is tested to the satisfaction of business customers and service provider staff;

e) 确保适当的发布授权人员签署每一阶段的验收测试；

e) ensure that the appropriate release authority signs off each stage of acceptance testing;

f) 在安装前，验证目标平台满足软

f) verify that the target platform satisfies the hardware and

硬件要求；

software prerequisites before installation;

- g) 当发布实现其目标时，验证发布的完整性。

- g) verify that a release is complete when it reaches its destination.

10.1.7. 文件 Documentation

应该依据已发布的配置项，建立和保存合适的文档，并按照配置管理过程管理。文件应该包括：

Appropriate documentation should be available on completion and stored under configuration management against the released configuration item. This documentation should include:

- a) 支持性文件，包括服务等级协议；

- a) support documentation e.g. service level agreements;

- b) 支持性文件，包括系统概述、安装和支持程序、辅助诊断、运行和管理指南；

- b) support documentation, e.g. system overview, installation and support procedures, diagnostic aids, operating and administration instructions;

- c) 编译、发布、安装和分发过程；

- c) build, release, installation and distribution processes;

- d) 持续性和回退计划；

- d) contingency and back-out plans;

- e) 服务管理人员、支持人员和客户的培训计划；

- e) training schedule for service management, support staff and customers;

- f) 发布的配置基准线，包括诸如系统文件、测试环境、测试文件、编译版本和开发工具等相关配置项；

- f) a configuration baseline for the release including associated configuration items such as system documentation, test environments, test documentation, versions of build and development tools;

- g) 相关变更、问题和已知错误；

- g) related changes, problems and known errors;

h) 关于发布授权、验证、接受的证据；	h) evidence of release authorization and related evidence of verification and acceptance.
一个系统或服务不能符合要求时，应该在导入运行系统前，通过配置管理和问题管理被识别并记录。	A system or service which does not completely conform to its specified requirements should be identified and recorded through configuration management and problem management prior to going live.
应该与事件管理交互已知错误的信息。	Information on known errors should be communicated to incident management.
如果发布被拒绝、延迟或取消，应告知变更管理人员。	If the release is rejected, delayed or cancelled, change management should be informed.
10.1.8. 回滚、分配和安装 Roll-out, distribution and installation	
应该对导入计划进行评审，并补充必要细节，以确保所有必要活动将被实施。	The roll out plan should be reviewed and detail added as necessary to ensure that all necessary activities will be performed.
关键之处在于，发布应该在预期状态下完成。回滚、发布和安装流程应确保：	It is important that the release is delivered safely to its destination in its expected state. The roll-out, distribution and installation processes should ensure that:
a) 所有硬件和软件存储区是安全的；	a) all hardware and software storage areas are secure;
b) 采用合适的程序管理物品的存储、分发、接收和销毁；	b) there are appropriate procedures for the storage, despatch, receipt and disposal of goods;
c) 计划并完成安装、环境准备、电	c) installation, environmental, electrical and facilities

子和设施的检查；	checks are planned and completed;
d) 向业务和服务提供方通报新的发布；	d) that business and service provider staff are notified of new releases;
e) 卸载多余的产品、服务和软件许可。	e) redundant products, services and licences are decommissioned.
当通过网络传送软件时，应该检查传送到目的地的软件是否完整和可运行。	After software distribution over a network it is essential to check that the release is complete and operational when it reaches its destination.
在成功完成安装之后，应该更新资产和配置管理的记录，包括软硬件的位置和负责人。	After a successful installation the asset and configuration management records should be updated with the location and the owner of the hardware and software.
可以采用客户安装接受和满意度问卷来检查安装是否成功。客户的反馈应该被及时反馈给业务关系管理过程。	An installation customer acceptance and satisfaction questionnaire may be used to record success or failure. Results of any customer surveys should be fed back to business relationship management.
10.1.9. 发布后处理和回退 Post release and roll-out	
在发布导入后一段时间内，与发布相关的应该对事件数量进行测量和分析，以评估对业务业务、运营和支持人员的影响。	The number of incidents related to the release in the period immediately following a roll-out should be measured and analysed to assess their impact on the business, operations and support staff resources.
变更管理流程应包括实施后的评审。	The change management process should include a post-implementation review.
相关建议应该作为服务改进计划的	Recommendations should be fed into a plan for improving

输入。

the service.

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